

EEG-Based Emotion Recognition via Channel-Wise Attention and Self Attention

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Abstract—Emotion recognition based on electroencephalography (EEG) is a significant task in the brain-computer interface field. Recently, many deep learning-based emotion recognition methods are demonstrated to outperform traditional methods. However, it remains challenging to extract discriminative features for EEG emotion recognition, and most methods ignore useful information in channel and time. This article proposes an attention-based convolutional recurrent neural network (ACRNN) to extract more discriminative features from EEG signals and improve the accuracy of emotion recognition. First, the proposed ACRNN adopts a channel-wise attention mechanism to adaptively assign the weights of different channels, and a CNN is employed to extract the spatial information of encoded EEG signals. Then, to explore the temporal information of EEG signals, extended self-attention is integrated into an RNN to recode the importance based on intrinsic similarity in EEG signals. We conducted extensive experiments on the DEAP and DREAMER databases. The experimental results demonstrate that the proposed ACRNN outperforms state-of-the-art methods.

Index Terms—Electroencephalogram (EEG), emotion recognition, channel-wise attention, self-attention

1 INTRODUCTION

EMOTION analysis is important in daily life, particularly in the human-computer interaction field [1], [2], [3], [4], [5]. Emotion analysis can help increase the quality of human-computer communication and improve the intelligence of computer. In addition, emotion analysis plays an important role in health care to understand the behavioral and cognitive functioning of patients [6], [7], and physiological signals are generally used to measure the emotional state, including galvanic skin response, electromyography, heart rate, respiration rate and electroencephalography (EEG) [8].

In the past decade, the relationship between emotion and EEG signals has been studied extensively [9], [10], [11]. EEG signals, which can be obtained easily, measure voltage fluctuations resulting from ionic current flows in the neurons of the brain [12]. EEG is noninvasive, fast, and inexpensive,

thereby making it a preferred method to brain responses to emotional stimuli. In addition, EEG signals are widely used for emotion analysis because EEG can be explored various information about emotions from frequency band, electrode position, and temporal information [13], [14], [15].

Generally, most EEG emotion recognition methods first design features from EEG signals and adopt classifiers to classify the emotion features. For example, Li *et al.* extracted features from the gamma frequency band and used a linear support vector machine (SVM) to classify the extracted features [13]. Patil *et al.* adopted higher-order crossings as features, which are better than other statistical features to classify emotions [16]. Shi *et al.* first proposed differential entropy (DE) features from five frequency bands and validated that DE features are superior for representing EEG signals [17]. In addition, Duan *et al.* extracted DE features from multichannel EEG data and combined an SVM and k-Nearest Neighbor (KNN) to classify the extracted features [18].

Recently, deep learning has been demonstrated to outperform traditional machine learning in many fields, e.g., computer vision [19], natural language processing [20] and biomedical signal processing [21], [22], [23]. In addition, many deep learning-based methods have been widely used for EEG-based emotion recognition. On one hand, deep learning methods can be considered as classifiers after feature extraction. For example, Yang *et al.* combined the DE of multiple bands as EEG features and employed a continuous convolutional neural network as a classifier [24]. Song *et al.* designed DE features according to the electrode position relationship and adopted a graph convolutional neural network as a classifier [25]. On the other hand, many deep learning methods are data-driven and function in an end-to-end manner, which does not require handcrafted features from EEG signals. For example, Alhagry *et al.* proposed an end-to-end deep learning neural network to recognize emotion from raw EEG signals, which used an LSTM-RNN to learn features from EEG signals and used the dense layer for

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基于脑电图的情绪识别：通道级注意力和自注意力

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摘要—基于脑电图（EEG）的情绪识别是脑机接口领域的一项重要任务。近年来，许多基于深度学习的情绪识别方法被证明优于传统方法。然而，提取 EEG 情绪识别的判别性特征仍然具有挑战性，并且大多数方法忽略了通道和时间中的有用信息。本文提出了一种基于注意力的卷积循环神经网络（ACRNN），用于从 EEG 信号中提取更具判别性的特征，并提高情绪识别的准确性。首先，所提出的 ACRNN 采用通道注意力机制来自适应地分配不同通道的权重，并使用 CNN 提取编码 EEG 信号的空间信息。然后，为了探索 EEG 信号的时间信息，扩展自注意力机制被集成到 RNN 中，以基于 EEG 信号中的内在相似性重新编码重要性。我们在 DEAP 和 DREAMER 数据库上进行了广泛的实验。实验结果表明，所提出的 ACRNN 优于现有最佳方法。

索引词—脑电图 (EEG)、情绪识别、通道注意力、自注意力



1 引言

情绪分析在日常生活中的重要性日益凸显，特别是在人机交互领域 [1]、[2]、[3]、[4]、[5]。情绪分析有助于提升人机沟通的质量，并增强计算机的智能化。此外，情绪分析在医疗保健领域也发挥着重要作用，有助于理解患者的行为和认知功能 [6]、[7]，而生理信号通常被用于测量情绪状态，包括皮肤电反应、肌电图、心率、呼吸率和脑电图 (EEG) [8]。在过去十年中，情绪与脑电图信号之间的关系得到了广泛研究 [9]、[10]、[11]。脑电图信号易于获取，能够测量大脑神经元离子电流流动产生的电压波动 [12]。脑电图具有无创性、快速、低成本的特点，

因此，它成为评估大脑对情绪刺激反应的首选方法。此外，脑电图信号广泛应用于情绪分析，因为脑电图可以从频段、电极位置和时间信息中探索各种情绪信息 [13]，[14]，[15]。

通常，大多数脑电图情绪识别方法首先从脑电图信号中提取特征，并采用分类器对情绪特征进行分类。例如，Li 等人从伽马频段提取特征，并使用线性支持向量机（SVM）对提取的特征进行分类 [13]。Patil 等人采用高阶穿越作为特征，这些特征比其他统计特征更适合分类情绪 [16]。Shi 等人首先从五个频段提出了差分熵（DE）特征，并验证了 DE 特征在表示脑电图信号方面的优越性 [17]。此外，Duan 等人从多通道脑电图数据中提取 DE 特征，并结合 SVM 和 k-近邻（KNN）对提取的特征进行分类 [18]。

近年来，深度学习在许多领域已被证明优于传统机器学习，例如计算机视觉 [19]、自然语言处理 [20] 和生物医学信号处理 [21]、[22]、[23]。此外，许多基于深度学习的方法已被广泛应用于脑电图

（EEG）情感识别。一方面，深度学习方法在特征提取后可被视为分类器。例如，杨等人将多个频带的 DE 结合作为 EEG 特征，并采用连续卷积神经网络作为分类器 [24]。宋等人根据电极位置关系设计 DE 特征，并采用图卷积神经网络作为分类器 [25]。另一方面，许多深度学习方法是基于数据的，并以端到端的方式运行，这不需要从 EEG 信号中提取手工特征。例如，Alhagry 等人提出了一种端到端深度学习神经网络，用于从原始 EEG 信号中识别情感，该网络使用 LSTM-RNN 从 EEG 信号中学习特征，并使用密集层进行

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classification [15]. Yang *et al.* proposed a parallel convolutional recurrent neural network for EEG emotion recognition and achieved good performance [7]. However, it still remains challenging to extract more discriminative features for EEG emotion recognition. Therefore, it is important to design an effective deep learning framework that can extract features and perform classification directly from raw EEG signals.

Inspired by the cascade convolutional recurrent network (CRNN), which combines CNN and RNN to extract spatial and temporal features from EEG signals [26], we use a CNN to extract the spatial information of EEG signals. Then, we employ two long short-term memory (LSTM) layers to extract temporal information, which is better at storing and accessing information than a standard RNN [27]. Different from a traditional CRNN, we employ a framework to extract more discriminative spatiotemporal information using two attention mechanisms, i.e., a channel-wise attention mechanism [28] and an extended self-attention mechanism [29]. Generally, CNNs are used to extract the spatial features of EEG signals [7], however, this ignores the importance of features among different channels. To extract more discriminative features from the spatial information, some methods adopt channel selection to choose more relevant channels [30]. Different from traditional methods that need first select the relevant channels artificially [31], in this study, we first adopt an adaptive channel-wise mechanism, that transforms channels to a probability distribution as weights and recodes the EEG signals based on the transformed weights. Then CNNs are employed to extract the discriminative spatial features of recoded signals. In addition, an RNN is employed to explore the time information of EEG signals, however, this also ignores the importance of different EEG samples. Note that extended self-attention can be applied to LSTM to utilize long-range dependencies [32]. We integrate the extended self-attention mechanism into the RNN to explore the importance of different EEG samples, because this mechanism can update the weight according to the similarity of EEG signals. As a result, more discriminative temporal and spatial characteristics of EEG signals can be obtained by integrating the two attention mechanisms in our framework.

In this paper, we propose the attention-based convolutional recurrent neural network (ACRNN) to deal with EEG-based emotion recognition. Raw EEG signals can contain spatial information by the intrinsic relationship among different channels and time dependence among temporal slices, thus, the proposed ACRNN can learn the spatial features of multichannel EEG in the convolutional layer and explore the temporal features of different temporal slices using LSTM networks. In addition, the channel-wise attention and extended self-attention mechanisms can extract more discriminative spatial and temporal features, respectively. The proposed model was evaluated on two publicly available databases, i.e., DEAP [2] and DREAMER [3], and the proposed method demonstrated superior performance relative to recognition accuracy in two databases. Our primary contributions are summarized as follows.

1) We have developed a data-driven ACRNN framework for EEG-based emotion recognition. This framework integrates the channel-wise attention mechanism into a CNN to explore spatial information, which can take the importance of

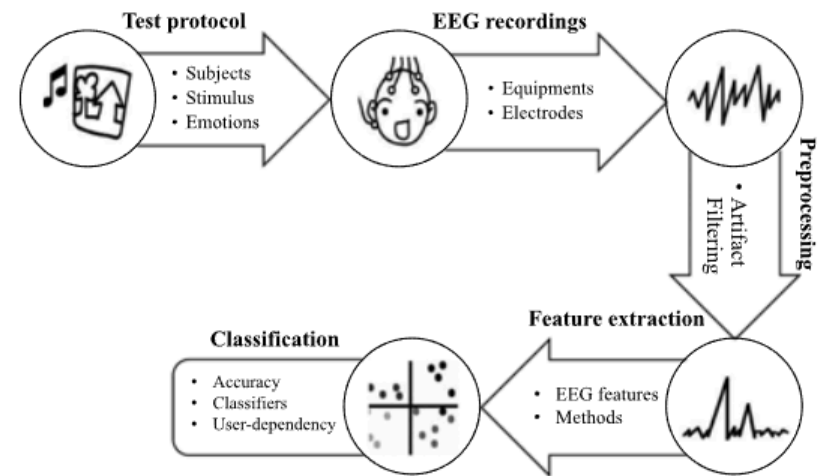


Fig. 1. The general flow of EEG emotion recognition.

different channels by channel-wise attention and the spatial information of multichannel EEG signals by a CNN into consideration. Besides, ACRNN integrates extended self-attention mechanism into RNN to explore temporal information of EEG signals, which can take the different temporal information by LSTM and the intrinsic similarity of each EEG sample by extended self-attention into consideration.

2) We conducted experiments on the DEAP and DREAMER databases, and the experimental results indicate average emotion recognition accuracies of 92.74 and 93.14 percent in the valence and arousal classification tasks of the DEAP database, respectively. In addition, the proposed method achieved mean accuracies of 97.79, 97.98 and 97.67 percent in the valence, arousal and dominance classification tasks of the DREAMER database, respectively.

The remainder of this paper is organized as follows. Section 2 introduces related work, and Section 3 presents the proposed method. Section 4 discusses extensive experiments conducted to demonstrate the effectiveness of the proposed ACRNN. Finally, a discussion is given in Section 5, and the paper is concluded in Section 6.

2 RELATED WORK

Here, we introduce the general flow of the traditional EEG emotion recognition framework. We then introduce the channel-wise attention and self-attention mechanisms.

2.1 General Flow of EEG Emotion Recognition

Recently, emotion recognition from EEG signals has received significant attention. The general flow of the EEG emotion recognition framework is summarized as follows (Fig. 1).

- i) Test protocol: First, the type of stimulus used, trial duration, the number of subjects, their gender, and the emotions to be recognized are recorded. Then, the subjects are exposed to the stimulus, e.g., music or a movie [2], [3].
- ii) EEG recordings: The number of electrodes and test duration are recorded, and then EEG signals are recorded by electrodes. The subjects then assess their emotional state by labeling the EEG recording after each trial [2], [3].
- iii) Preprocessing: To avoid artifacts in the EEG signals, e.g., eye blinks, the EEG signals should be preprocessed using artifact removal methods, e.g., blind source separation and independent component analysis [33].

分类[15]。杨等人提出了一种用于脑电图情绪识别的并行卷积循环神经网络，并取得了良好的性能[7]。然而，提取脑电图情绪识别的更具判别性的特征仍然是一个挑战。因此，设计一个能够直接从原始脑电图信号中提取特征并执行分类的有效深度学习框架非常重要。

受级联卷积循环网络（CRNN）的启发，该网络结合 CNN 和 RNN 从脑电图（EEG）信号中提取空间和时间特征[26]，我们使用 CNN 提取 EEG 信号的空间信息。然后，我们采用两个长短期记忆

（LSTM）层提取时间信息，这比标准 RNN 更擅长存储和访问信息[27]。与传统的 CRNN 不同，我们采用一个框架，通过两种注意力机制提取更具判别性的时空信息，即通道注意力机制[28]和扩展自注意力机制[29]。通常，CNN 用于提取 EEG 信号的空间特征[7]，但这忽略了不同通道间特征的重要性。为了从空间信息中提取更具判别性的特征，一些方法采用通道选择来选择更相关的通道[30]。与需要人工首先选择相关通道的传统方法不同[31]，在本研究中，我们首先采用一种自适应的通道机制，将通道转换为概率分布作为权重，并根据转换后的权重重新编码脑电图信号。然后使用卷积神经网络提取重新编码信号的判别性空间特征。此外，使用循环神经网络探索脑电图信号的时间信息，然而，这也忽略了不同脑电图样本的重要性。请注意，扩展自注意力机制可以应用于 LSTM 以利用长距离依赖关系[32]。我们将扩展自注意力机制集成到循环神经网络中，以探索不同脑电图样本的重要性，因为这种机制可以根据脑电图信号的相似性更新权重。因此，通过在我们的框架中集成这两种注意力机制，可以获得更具判别性的脑电图信号的时间和空间特征。

在本文中，我们提出了基于注意力的卷积循环神经网络（ACRNN）来处理基于脑电图（EEG）的情绪识别。原始的脑电图信号可以通过不同通道之间的内在关系包含空间信息，以及时间切片之间的时间依赖性，因此，所提出的 ACRNN 可以在卷积层中学习多通道脑电图的空间特征，并使用 LSTM 网络探索不同时间切片的时间特征。此外，通道注意力和扩展自注意力机制可以分别提取更具判别性的空间和时间特征。所提出的模型在两个公开可用的数据库，即 DEAP [2] 和 DREAMER [3] 上进行了评估，并且所提出的方法在两个数据库的识别准确率方面表现出优越的性能。我们的主要贡献总结如下。

1) 我们开发了一个用于基于脑电图的情绪识别的数据驱动 ACRNN 框架。该框架将通道注意力机制集成到 CNN 中，以探索空间信息

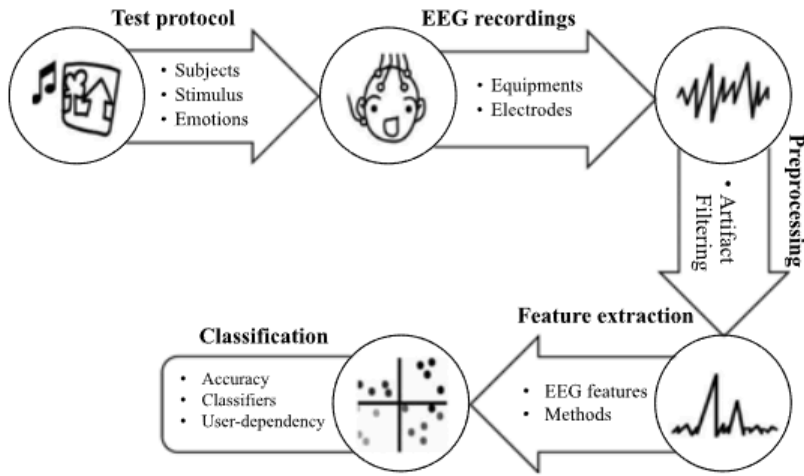


图 1. EEG 情绪识别的一般流程。

能够通过通道注意力机制考虑不同通道的重要性，并通过 CNN 考虑多通道脑电信号的时空信息。此外，ACRNN 将扩展自注意力机制集成到 RNN 中，以探索脑电信号的时序信息，能够考虑 LSTM 的不同时序信息和扩展自注意力机制中每个脑电样本的内在相似性。

我们在 DEAP 和 DREAMER 数据库上进行了实验，实验结果表明，在 DEAP 数据库的效价和唤醒度分类任务中，平均情绪识别准确率分别为 92.74%和 93.14%。此外，所提出的方法在 DREAMER 数据库的效价、唤醒度和支配度分类任务中分别达到了 97.79%、97.98% 和 97.67%的平均准确率。

本文的其余部分组织如下。

第 2 节介绍相关工作，第 3 节介绍所提出的方法。第 4 节讨论了为验证所提出 ACRNN 有效性的广泛实验。最后，第 5 节给出讨论，第 6 节总结全文。

2 相关工作

这里，我们介绍了传统 EEG 情绪识别框架的一般流程。然后，我们介绍了通道注意力机制和自注意力机制。

2.1 EEG 情绪识别的一般流程

近年来，基于 EEG 信号的情绪识别受到了广泛关注。EEG 情绪识别框架的一般流程总结如下（图 1）。

- i) 测试协议：首先，记录所使用的刺激类型、试验持续时间、受试者数量、性别以及需要识别的情绪。然后，让受试者接触刺激，例如音乐或电影[2]，[3]。
- ii) EEG 记录：电极数量和测试持续时间被记录下来，然后通过电极记录脑电图信号。受试者在每次试验后通过标记脑电图记录来评估他们的情绪状态[2]，[3]。
- iii) 预处理：为了避免脑电图信号中的伪影，例如，眼动时，EEG 信号应使用伪影去除方法进行预处理，例如盲源分离和独立成分分析[33]。

- iv) Feature extraction: To extract relevant emotion features from EEG signals, information about the EEG signals is explored, e.g., the EEG characteristics in the time, frequency, and spatial domains [9].
- v) Various classifiers can be used to classify the extracted features, e.g., Bayesian, support vector machines, decision trees, and deep learning classifiers [34]. Depending on whether the classifier was trained on user-dependent data, EEG emotion recognition can be also divided into user-dependent and user-independent tasks.

2.2 Channel-Wise Attention

Attention plays an important role in human perception [35], [36]. For example, humans can exploit a sequence of partial glimpses and selectively focus on salient parts to better capture a visual structure [37]. Inspired by the human attention mechanism, spatial attention mechanisms have been proposed for various vision tasks, e.g., semantic attention [38], multi-layer attention [39] and channel-wise attention [28]. Channel-wise attention demonstrates superior performance because it can change the weight of different channels to explore the information of a feature map; thus, it can extract more important information about channels. Therefore, the channel-wise attention mechanism has been used to exploit interdependencies among feature channels. For example, Hu *et al.* introduced a compact module to exploit the inter-channel relationship of a feature map [40], and Chen *et al.* combined spatial and channel-wise attention for image captioning [28].

Generally, channel-wise attention can squeeze the global spatial information and generate channel-wise statistics [28]. In addition, it is trainable with CNNs, thus, it can be integrated into CNN architectures [41]. Considering that multichannel EEG signals contains the spatial information via channels, channel-wise attention can be integrated into a CNN to explore the importance among the channels of EEG signals, and more discriminative spatial information can be extracted by a CNN.

2.3 Self Attention

Self-attention is an intra-attention mechanism that relates different positions of a single sequence to encode sequence data based on an importance score [20]. In addition, the self-attention mechanism is popular because it can improve long-range dependency modeling [42]. An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. Self-attention has been demonstrated to perform well on simple-language question answering and language modeling tasks. For example, Vaswani *et al.* proposed an attention-based architecture for machine translation [32], and Shen *et al.* proposed the directional self-attention network to focus on the attention among elements in an input sequence [29]. In EEG recognition tasks, to augment the amount of training samples, one EEG trial is often segmented into several input samples. However, many methods ignore

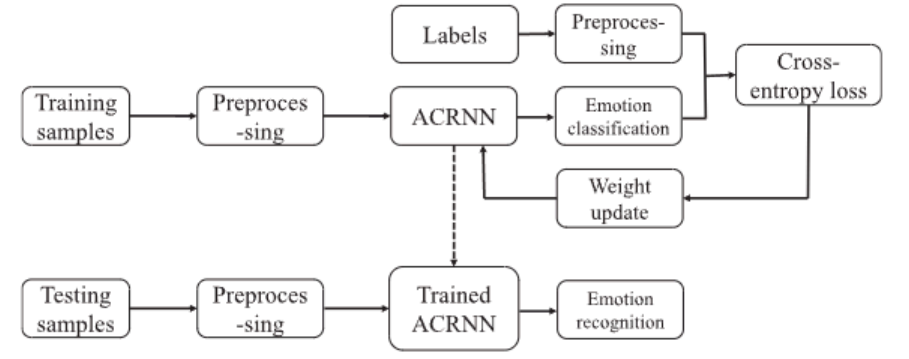


Fig. 2. Overview of the attention based convolutional recurrent neural network on EEG-based emotion recognition.

the importance of different EEG samples. Inspired by self-attention, we adopt this technique to further explore the time dependence among EEG samples.

3 PROPOSED METHODS

In this section, we first introduce the proposed EEG emotion recognition framework, and then we introduce our raw EEG signal preprocessing technique. Finally, we describe the construction of the proposed ACRNN in detail.

3.1 Framework of Proposed ACRNN

Generally, most EEG-based emotion recognition studies have focused on first extracting relevant features, and then the extracted features are used to classify the subject's emotional state [8], [14], [25]. In practice, raw EEG signals contain rich spatial and temporal information, which can be extracted to recognize a subject's emotional state. The proposed ACRNN is a data-driven method that integrates channel-wise [28] and extended self-attention [29] mechanisms into a CNN-RNN simultaneously. In addition, ACRNN can extract spatial and temporal information as emotion features, and classifies the extracted features using softmax function. Consequently, this end-to-end technique improves the accuracy of EEG based emotion recognition (Fig. 2). First, we divide the EEG samples into training and testing samples. Then, the training and testing samples are preprocessed by removing baseline signals, respectively. In addition, labels are preprocessed using the slicing window technique. Next, we use the training samples to train the proposed ACRNN model, compute the cross-entropy loss and update network parameters using the Adam optimizer [43]. Finally, the trained model is used to identify the emotional state of the testing samples, and classification accuracy are considered as the final recognition results.

3.2 Preprocessing of Proposed ACRNN

In the proposed ACRNN, preprocessing involves removing baseline signals and sliding windows. Generally, recorded EEG signals contain baseline and trial signals [2], [3]. Yang *et al.* proposed that baseline removal preprocessing can improve EEG emotion recognition on the DEAP database [7]. Here, let $\mathbf{X}_R = [\mathbf{X}_B, \mathbf{X}_T] \in \mathbb{R}^{M \times N}$ be the recorded EEG signals with H Hz sampling frequency and duration T_1 , where M is the number of EEG electrode nodes, N is the number of sampling points. In addition, $\mathbf{X}_B \in \mathbb{R}^{M \times L}$ denotes the baseline signal with duration T_2 , L denotes the number of sampling points, $X_i (i = 1, 2, \dots, T_2) \in \mathbb{R}^{M \times H}$ denotes the i th second baseline signals. Thus, the mean value of baseline signal per-second can be formulated as follows:

- iv) 特征提取：为了提取相关的情绪特征
从脑电图信号中提取特征，探索脑电图信号的信息，例如时域、频域和空间域的脑电图特征[9]。
- v) 可以使用各种分类器对提取的特征进行分类，例如贝叶斯分类器、支持向量机、决策树和深度学习分类器[34]。根据分类器是否在用户相关数据上训练，脑电图情绪识别也可以分为用户相关和用户无关的任务。

2.2 通道注意力

注意力在人类感知中扮演着重要角色[35], [36]。例如，人类可以利用一系列部分视野，并选择性地聚焦于显著部分，以更好地捕捉视觉结构[37]。受人类注意力机制启发，空间注意力机制已被提出用于各种视觉任务，例如语义注意力[38]、多层注意力[39]和通道注意力[28]。通道注意力表现出更优越的性能，因为它可以改变不同通道的权重以探索特征图的信息；因此，它可以提取更多关于通道的重要信息。因此，通道注意力机制被用于利用特征通道之间的相互依赖关系。例如，Hu 等人引入了一个紧凑模块来利用特征图的内通道关系[40]，Chen 等人将空间注意力和通道注意力结合起来用于图像描述[28]。通常情况下，通道注意力机制可以压缩全局空间信息并生成通道统计信息[28]。此外，它可以与 CNN 一起训练，因此可以集成到 CNN 架构中[41]。考虑到多通道脑电图信号通过通道包含空间信息，通道注意力机制可以集成到 CNN 中，以探索脑电图信号通道间的重要性，CNN 可以提取更具判别性的空间信息。

2.3 自注意力

自注意力是一种内部注意力机制，它将单个序列的不同位置关联起来，根据重要性分数对序列数据进行编码[20]。此外，自注意力机制因其能够改进长距离依赖建模[42]而广受欢迎。注意力函数可以描述为将查询和一组键值对映射到输出，其中查询、键、值和输出都是向量。输出计算为值的加权求和，其中分配给每个值的权重由查询与相应键的兼容性函数计算得出。自注意力已在简单的语言问答和语言建模任务中表现出色。例如，Vaswani 等人提出了一种基于注意力的机器翻译架构[32]，Shen 等人提出了定向自注意力网络，以专注于输入序列中元素之间的注意力[29]。在脑电图识别任务中，为了增加训练样本的数量，一个脑电图试验通常被分割成多个输入样本。然而

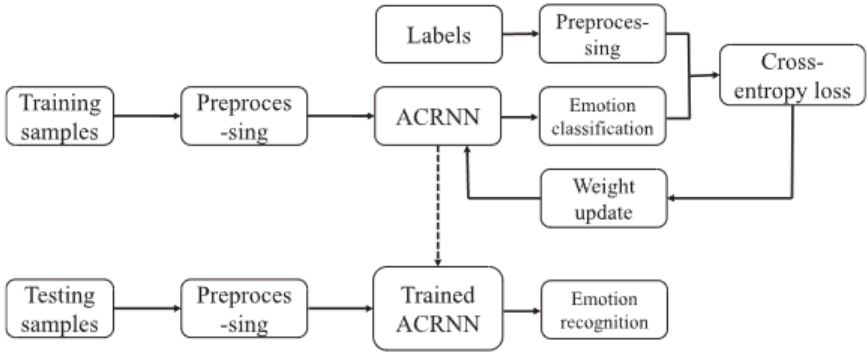


图 2. 基于脑电图的情绪识别的注意力卷积循环神经网络概述。

许多方法忽略了不同脑电图样本的重要性。受自注意力机制启发，我们采用这一技术进一步探索脑电图样本之间的时序依赖性。

3 拟议方法

在本节中，我们首先介绍拟议的脑电图情绪识别框架，然后介绍我们的原始脑电图信号预处理技术。最后，我们详细描述拟议的 ACRNN 的构建。

3.1 拟议 ACRNN 框架

通常情况下，大多数基于脑电图（EEG）的情绪识别研究首先提取相关特征，然后使用提取的特征来分类被试的情绪状态[8], [14], [25]。在实践中，原始的脑电图信号包含丰富的空间和时间信息，这些信息可以被提取出来以识别被试的情绪状态。所提出的 ACRNN 是一种数据驱动方法，将通道级[28]和扩展自注意力[29]机制同时集成到 CNN-RNN 中。此外，ACRNN 可以提取空间和时间信息作为情绪特征，并使用 softmax 函数对提取的特征进行分类。因此，这种端到端技术提高了基于脑电图的情绪识别的准确性（图 2）。首先，我们将脑电图样本分为训练样本和测试样本。然后，分别通过去除基线信号对训练样本和测试样本进行预处理。此外，使用切片窗口技术对标签进行预处理。接下来，我们使用训练样本来训练所提出的 ACRNN 模型，并使用 Adam 优化器[43]计算交叉熵损失并更新网络参数。最后，使用训练好的模型来识别测试样本的情绪状态，并将分类准确率作为最终的识别结果。

3.2 提出 ACRNN 的预处理

在提出的 ACRNN 中，预处理包括去除基线信号和滑动窗口。通常，记录的脑电图（EEG）信号包含基线和试验信号[2], [3]。Yang 等人提出基线去除预处理可以提高 DEAP 数据库上的脑电图情绪识别[7]。这里，设 $X \in \mathbb{R}^{M \times N}$ 为采样频率为 H Hz、持续时间为 T 的记录脑电图信号，其中 M 是脑电图电极节点数量， N 是采样点数量。此外， $X_0 \in \mathbb{R}^{M \times L}$ 表示持续时间为 T 的基线信号， L 表示采样点数量， $X_{0i} \in \mathbb{R}^{M \times L}$ 表示第 i 秒的基线信号。因此，每秒基线信号的平均值可以表示如下：

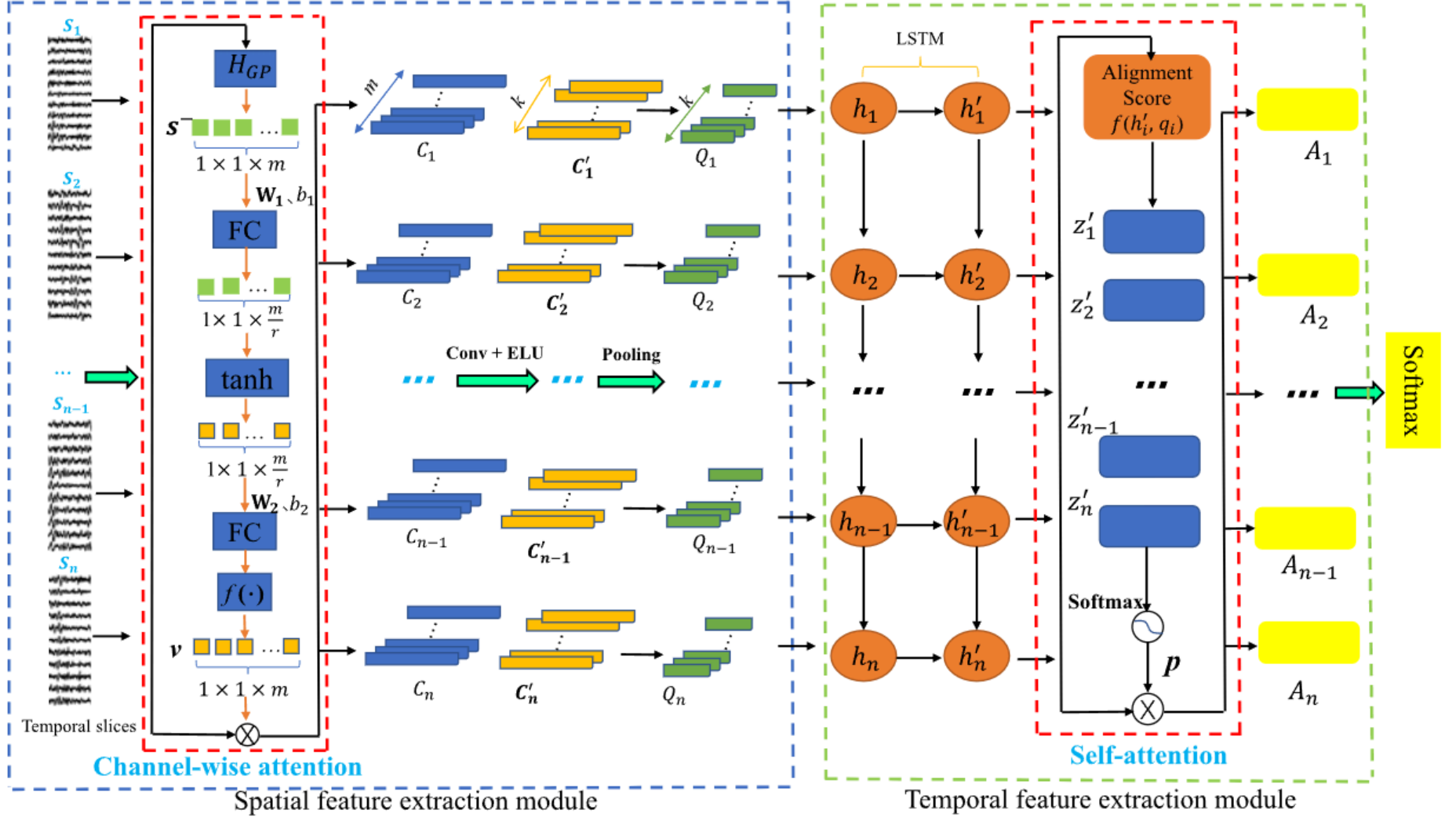


Fig. 3. The structure diagram of the attention based convolutional recurrent neural network.

$$\bar{\mathbf{X}}_{\mathbf{B}} = \frac{\sum_{i=1}^{T_2} \mathbf{X}_i}{T_2}, \quad (1)$$

where $\bar{\mathbf{X}}_{\mathbf{B}} \in \mathbb{R}^{M \times H}$ denotes the mean value of baseline signal per-second. Let $\mathbf{X}_{\mathbf{T}} \in \mathbb{R}^{M \times J}$ represents the trial EEG signals with duration T_3 , where J is the number of sampling points. To remove the baseline of trial EEG signals, $\mathbf{X}_{\mathbf{T}}$ is segmented into several slices $\mathbf{X}_j$ ($j = 1, 2, \dots, T_3$) $\in \mathbb{R}^{M \times H}$ with a one-second nonoverlapping sliding window, and the per-second baseline-removed signals can be formulated as follows:

$$\mathbf{X}'_j = \mathbf{X}_j - \bar{\mathbf{X}}_{\mathbf{B}}. \quad (2)$$

Finally, these baseline signals removed temporal slices $\mathbf{X}'_j$ are concatenated into a new matrix $\mathbf{X}_{\mathbf{T}}' \in \mathbb{R}^{M \times J}$.

Generally, to augment the amount of training data, one EEG trial $\mathbf{X}_{\mathbf{T}}' \in \mathbb{R}^{M \times J}$ is often segmented into several temporal slices $\mathbf{S} = \{\mathbf{S}_1, \mathbf{S}_2, \dots, \mathbf{S}_n\}$ by sliding window. Here, $\mathbf{S}_i$ ($i = 1, 2, \dots, n$) $\in \mathbb{R}^{M \times T}$ represents the i th EEG sample, and T denotes the number of sampling points in each sliding window. Generally, a human emotional state lasts from 1 s to 12 s, and previous studies have shown that a 3-s sliding window can achieve good classification accuracy [44], thus, we employ a 3-s sliding window, i.e., $T/H = 3$.

3.3 The Construction of Proposed ACRNN

The proposed ACRNN comprises the channel-wise attention mechanism, a CNN, an RNN and the extended self-attention mechanism. The structure of the proposed ACRNN is shown in Fig. 3. The left side of diagram shows the spatial feature extraction module. First, to explore the importance among the different channels of multichannel EEG signals, we employ the attention mechanism in a channel-wise manner into the EEG signals (Fig. 3). In actual EEG signal acquisition, different EEG channels in multichannel devices often contain

redundant or less relevant information. To enhance emotion recognition accuracy, some methods adopt channel selection to choose more relevant channels [30]. Different from traditional methods that need select relevant channels artificially [31], we adopt the adaptive channel-wise mechanism, which can consider the information of all channels and assign weights to different channels based on importance. In our framework, $\mathbf{S} = \{\mathbf{S}_1, \mathbf{S}_2, \dots, \mathbf{S}_n\}$ represents EEG samples after preprocessing, and $\mathbf{S}_i = [s_1, s_2, \dots, s_m]$ ($i = 1, 2, \dots, n$) is the i th EEG sample, where s_j ($j = 1, 2, \dots, m$) represents the j th channel of EEG sample $\mathbf{S}_i$, and m is the total number of channels of each sample. In this model, we first apply mean pooling for each channel of EEG sample to obtain channel-wise statistics as follows:

$$\mathbf{s}^- = [s_1^-, s_2^-, \dots, s_m^-], \quad (3)$$

where s_j^- ($j = 1, 2, \dots, m$) is the mean of the j th channel. To reduce model complexity and improve generalizability, the channel-wise attention mechanism adopts two fully-connected (FC) layers around the non-linearity, i.e., a dimensionality-reduction layer with parameter $\mathbf{W}_1$ and bias terms b_1 with reduction ratio r and $\tanh$ function as the activation function, and a dimensionality increasing layer with parameter $\mathbf{W}_2$ and bias terms b_2 . Thus, the gating mechanism of channel-wise attention is expressed as follows:

$$\mathbf{v} = \text{softmax}(\mathbf{W}_2 \cdot (\tanh(\mathbf{W}_1 \cdot \mathbf{s}^- + b_1) + b_2)), \quad (4)$$

where the softmax function transforms the importance of channels to probability distribution $\mathbf{v} = [v_1, v_2, \dots, v_m]$, which represents the importance of different channels. Finally, we consider probability as the weight to recode the information of the EEG sample $\mathbf{S}_i = [s_1, s_2, \dots, s_m]$ in each channel. Thus, the j th ($j = 1, 2, \dots, m$) attentive channel feature extracted via channel-wise attention can be represented as follows:

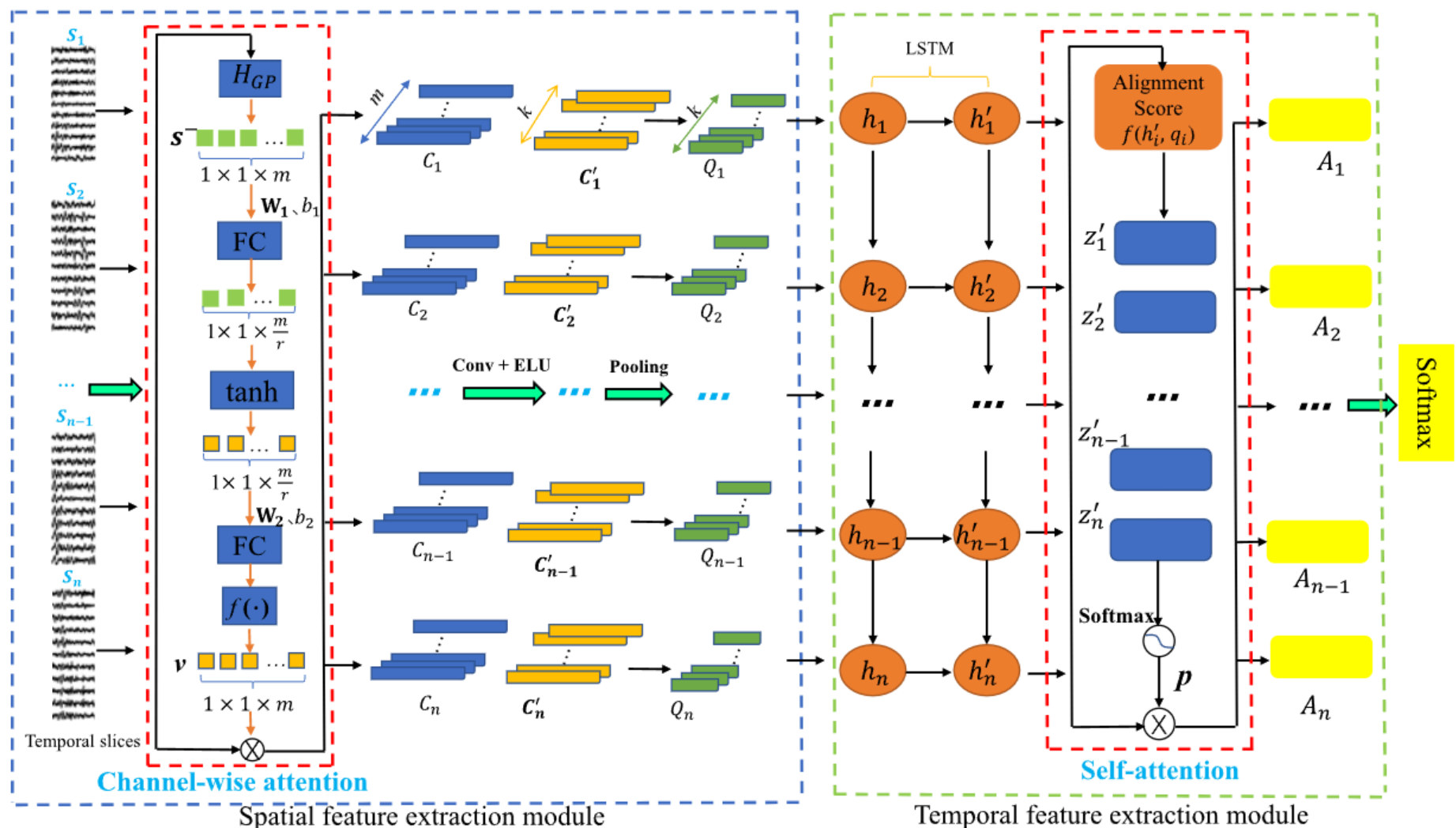


图 3. 基于注意力的卷积循环神经网络的结构图。

$$\bar{X}^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots \quad (1)$$

其中 $\bar{X}^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots$ 表示每秒基线信号的平均值。 $\bar{X}^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots$ 表示持续时间为 T 的试验脑电信号，其中 J 是采样点的数量。为了去除试验脑电信号的基线， X 被分割成多个 $X^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots$ 片段，使用一个一秒的非重叠滑动窗口，每秒的基线去除信号可以表示如下：

$$\bar{X}^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots \quad (2)$$

最后，这些去除基线的时序片段 X 被连接成一个新的矩阵 $\bar{X}^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots$ 。通常，为了增加训练数据的数量，一个脑电试验 $\bar{X}^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots$ 通常通过滑动窗口分割成多个时序片段 $S_1, S_2, \dots, S_g$ 。这里， $S_i^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots$ 表示第 i 个脑电样本， T 表示每个滑动窗口中的采样点数量。通常，人类情绪状态持续时间为 1 秒到 12 秒，先前研究表明，3 秒的滑动窗口可以实现良好的分类精度 [44]，因此，我们采用 3 秒的滑动窗口，即 $T = H^{1/4} \times 3$ 。

3.3 所提出的 ACRNN 的构建

所提出的 ACRNN 包括通道注意力机制、卷积神经网络、循环神经网络和扩展的自注意力机制。所提出的 ACRNN 的结构如图 3 所示。图的左侧显示了空间特征提取模块。首先，为了探索多通道脑电图信号中不同通道之间的重要性，我们将注意力机制以通道的方式应用于脑电图信号（图 3）。在实际脑电图信号采集

多通道设备中不同的脑电图（EEG）通道往往包含冗余或不那么相关的信息。为了提高情绪识别的准确性，一些方法采用通道选择来选择更相关的通道[30]。与我们需要人工选择相关通道的传统方法[31]不同，我们采用了自适应的通道机制，该机制可以考虑到所有通道的信息，并根据重要性为不同通道分配权重。在我们的框架中， $S = S_1, S_2, \dots, S_{\lfloor \frac{P}{T} \rfloor}$ 表示预处理后的 EEG 样本， $S = s_1, s_2, \dots, s_{\lfloor \frac{P}{T} \rfloor}$ ($i = 1, 2, \dots, m$) 是第 i 个 EEG 样本，其中 $s(j = 1, 2, \dots, m)$ 表示 EEG 样本 S 的第 j 个通道， m 是每个样本的总通道数。在这个模型中，我们首先对 EEG 样本的每个通道应用平均池化，以获得通道统计信息，如下所示：

$$S = S_1, S_2, \dots, s_{\lfloor \frac{P}{T} \rfloor} \quad (3)$$

其中 $s_j^{1/4} \times \frac{P}{T}^{i/4 \times 1} \times \dots$ 是第 j 个通道的平均值。为了降低模型复杂度并提高泛化能力，通道注意力机制在非线形周围采用了两个全连接（FC）层，即一个具有参数 W 和偏差项 b 的降维层，其降维比为 r ，激活函数为 $\tanh$ 函数，以及一个具有参数 W 和偏差项 b 的升维层。因此，通道注意力机制的门控机制表示如下：

$$v^{1/4} \times \text{softmax} \left(\frac{W \cdot \tanh(W \cdot s + b)}{P \cdot b + P} \right) \quad (4)$$

其中， softmax 函数将通道的重要性转换为概率分布 $v = [v_1; v_2; \dots; v_m]$ ，它表示不同通道的重要性。最后，我们将概率作为权重来重新编码每个通道中脑电图样本 $S = [s_1; s_2; \dots; s_m]$ 的信息。因此，通过通道注意力提取的第 j 个 ($j = 1, 2, \dots, m$) 注意力通道特征可以表示如下：

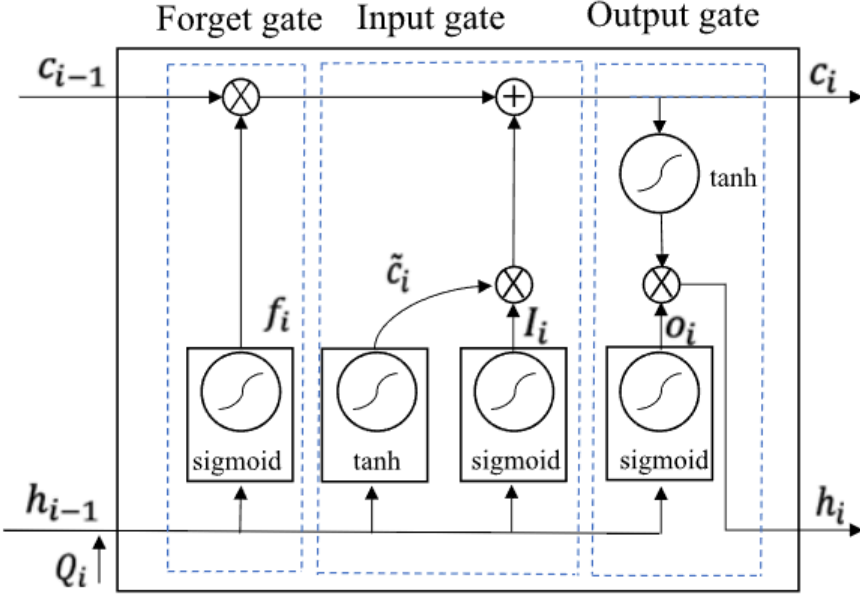


Fig. 4. LSTM unit architecture.

$$c_j = v_j \cdot s_j, \quad (5)$$

therefore, $\mathbf{C} = \{C_1, C_2, \dots, C_n\}$ represents the extracted channel-wise attentive features, the i th extracted feature $C_i = [c_1, c_2, \dots, c_m]$ can be obtained by channel-wise multiplication between each channel of $S_i = [s_1, s_2, \dots, s_m]$ and each element of $\mathbf{v} = [v_1, v_2, \dots, v_m]$.

Then, we use the CNN to further extract spatial information of EEG signals, where the number of convolution kernels is k , the kernel height is the same as the number of electrodes. Here, the kernel width is also designed to explore temporal information of the EEG signals. In addition, we use the exponential linear unit (ELU) function as the activation function in the convolution operations, which is better than the commonly used rectified linear unit (ReLU) function [45]. Thus, the i th feature C'_i ($i = 1, 2, \dots, n$) can be obtained from the i th channel attentive feature C_i after convolution and activation operations.

After that, we adopt a pooling layer to reduce the number of parameters and further extract features. Here, the i th encoded representation after pooling is $\{Q_i | Q_i = \text{MaxPool}(C'_i), i = 1 \dots n\}$.

The right side of the structure diagram shows the temporal feature extraction module (Fig. 3), which comprises a two-layer LSTM and extended self-attention mechanism. The LSTM network can learn the context information of the sequence because it is based on a recurrent structure [27]. The LSTM network has been successfully used for EEG emotion recognition because it can learn features from EEG data based on temporal dependence [15]. As shown in Fig. 4, an LSTM cell receives three inputs, i.e., input Q_i at the current time i , output c_{i-1} of previous time $i - 1$, and h_{i-1} representing the hidden state of the previous time $i - 1$. Then, the LSTM cell exports two outputs, i.e., output c_i at the current time i and hidden state h_i represented as the i th temporal feature extracted from LSTM. The LSTM cell contains three gates, i.e., the input, forget and output gates to control the data flow by the sigmoid and tanh activation functions. Although here training samples and testing samples are different and not consecutive in time, the encoded samples contain spatial information after the spatial feature extraction module. Meanwhile, the input gate weight is organized to learn spatial information while the forget weight map organizes to learn more temporal information, the input gate and forget gate compete with each

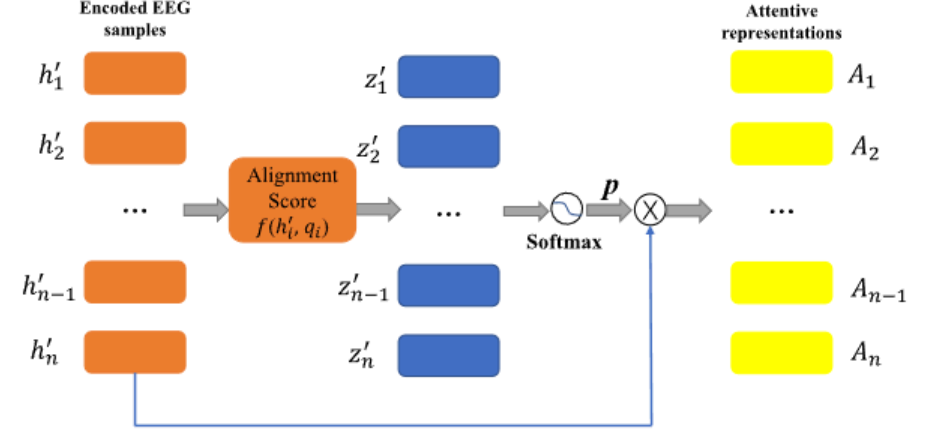


Fig. 5. Extended self-attention architecture.

other to input new information into the cell or keep the current temporal information, respectively [46]. Thus, the LSTM network can extract the spatiotemporal features.

In this study, the number of LSTM units in each layer is the same as the number of EEG samples, and the output in each time step can be considered as the temporal information extracted from each sample. Generally, the LSTM network adopts two stacked layers to remember and encode all scanned spatial and temporal areas [47], thus, we set the number of LSTM layers to two. Therefore, the i th output of the LSTM network is the hidden states of the second recurrent layer $\{h'_i | h'_i = \text{lstm}(Q_i), i = 1 \dots n\}$.

To extract more discriminative temporal information, we adopt the extended self-attention mechanism [29] to assign weights to each EEG signal sample by exploring the intrinsic importance of each sample. The structure of self-attention is shown in Fig. 5. Different from the traditional self-attention mechanism, which is used to assign importance to each recurrent encoded slice and aggregate this information to form a final representation [48], the extended attention is a natural extension of additive attention at the multi-dimensional feature level. It can better describe the specific meaning by computing the similarity within each sample from different points, and the obtained z'_i can be considered as a feature-wise score vector from the i th sample h'_i . In addition, the extended self-attention mechanism adds two bias terms to the inside and outside of the activation function, and the i th feature-wise score vector z'_i can be expressed as follows:

$$z'_i = f(h'_i, q_i) = W^T \sigma(W_1 h'_i + W_2 q_i + b_1) + b, \quad (6)$$

where $f(h'_i, q_i)$ represents the intrinsic similarity of the i th encoded EEG sample, and q_i is the aligned pattern vector generated based on the feature vector h'_i by linear transformation, where the dimension is the same as the feature vector. Here, the activation functions $\sigma(\cdot)$ is an exponential linear unit (ELU), W and b are the weight and bias terms of σ function, respectively, W_1, W_2 are weight parameters, and b_1 is the bias terms. Then, $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$ denotes the probabilities of all samples, and the probability of the i th EEG sample can be expressed as follows:

$$p_i = \frac{\exp(z'^T_i \cdot h'_i)}{\sum_{i=1}^n \exp(z'^T_i \cdot h'_i)}. \quad (7)$$

Lastly, $\mathbf{A} = \{A_1, A_2, \dots, A_n\}$ denotes the features extracted by the extended self-attention mechanism, and the i th attentive feature extracted by the extended self-attention mechanism can be computed as follows:

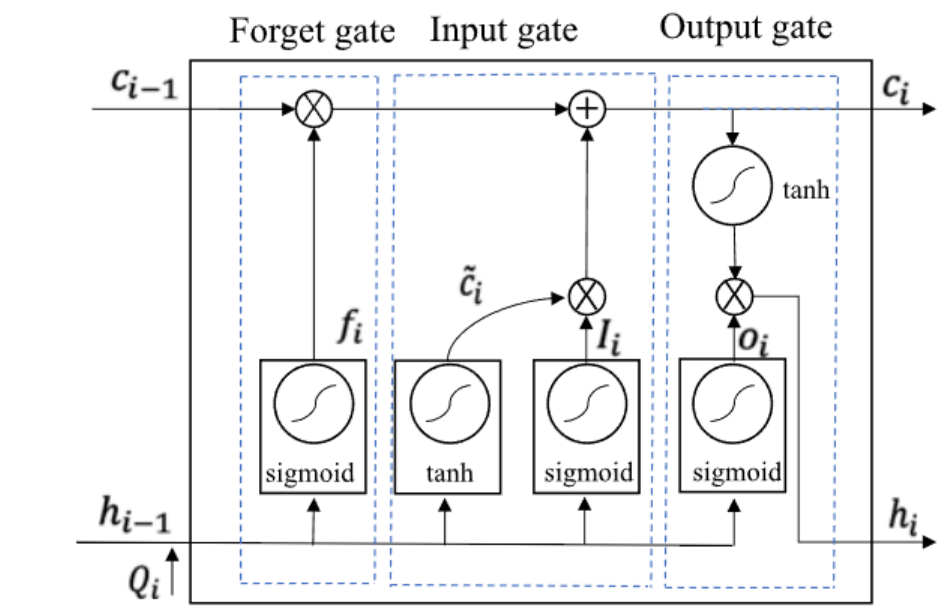


图 4. LSTM 单元架构。

$$C = V - S; \tag{5}$$

因此，提取的通道注意力特征，第 i 个提取特征 $C = (c; c; \dots; c)$ 可以通过 $S = (s; s; \dots; s)$ 的每个通道与 $V = (v; v; \dots; v)$ 的每个元素之间的通道乘法获得。

然后，我们使用 CNN 进一步提取脑电信号的时空信息，其中卷积核的数量为 k ，卷积核的高度与电极的数量相同。这里，卷积核的宽度也设计为探索脑电信号的时序信息。此外，我们在卷积操作中使用指数线性单元（ELU）作为激活函数，这比常用的整流线性单元（ReLU）函数[45]更好。因此，第 i 个特征 $C(i = 1; 2; \dots; n)$ 可以从卷积和激活操作后的第 i 个通道注意力特征 C 中获得。

之后，我们采用池化层来减少参数数量并进一步提取特征。这里，池化后的第 i 个编码表示为 $Q_j Q = \text{MaxPool } C$ ；

$i = 1 \dots ng$.

结构图右侧显示了时间特征提取模块（图 3），该模块包含两层 LSTM 和扩展的自注意力机制。LSTM 网络可以学习序列的上下文信息，因为它基于循环结构[27]。LSTM 网络已成功用于脑电图情绪识别，因为它可以根据时间依赖性从脑电图数据中学习特征[15]。如图 4 所示，LSTM 单元接收三个输入，即当前时间 i 的输入 Q 、前一时间 $i-1$ 的输出 c 和代表前一时间 $i-1$ 隐藏状态的 h 。然后，LSTM 单元输出两个输出，即当前时间 i 的输出 c 和表示从 LSTM 提取的第 i 个时间特征的隐藏状态 h 。LSTM 单元包含三个门，即输入门、遗忘门和输出门，通过 sigmoid 和 tanh 激活函数控制数据流。尽管这里的训练样本和测试样本不同且在时间上不连续，但经过空间特征提取模块编码后的样本包含空间信息。与此同时，输入门权重被组织起来以学习空间信息，而遗忘权重图则组织起来以学习更多的时间信息

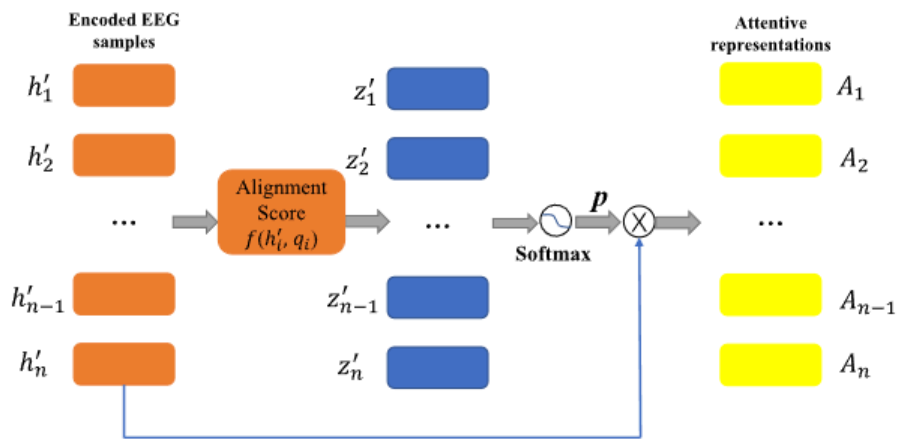


图 5. 扩展的自注意力架构。

输入门和遗忘门相互竞争，分别将新信息输入单元或保持当前时间信息[46]。因此，LSTM 网络可以提取时空特征。

在本研究中，每层中的 LSTM 单元数量与 EEG 样本数量相同，每个时间步的输出可以被视为从每个样本中提取的时间信息。通常，LSTM 网络采用两层堆叠来记忆和编码所有扫描的时空区域[47]，因此，我们将 LSTM 层数设置为两层。因此，LSTM 网络的第 i 个输出是第二层循环层的隐藏状态 $h_j = \text{lstm}(Q; i = 1 \dots n)$

为了提取更具判别性的时序信息，我们采用扩展的自注意力机制[29]来为每个 EEG 信号样本分配权重，通过探索每个样本的内在重要性。自注意力的结构如图 5 所示。与传统自注意力机制不同，后者用于为每个循环编码切片分配重要性并聚合这些信息以形成最终表示[48]，扩展注意力是多维特征层面的加性注意力的自然扩展。它可以通过计算不同点处每个样本内的相似性来更好地描述具体含义，获得的 z 可以视为来自第 i 个样本 h 的特征向量化得分向量。此外，扩展自注意力机制在激活函数的内部和外部添加了两个偏差项，第 i 个特征向量化得分向量 z 可以表示如下：

$$z = f(h; q) = \frac{1}{4} W_s W_h p W_q p b + b; \tag{6}$$

其中第 i 个编码的 EEG 样本的内在相似度， q 是基于特征向量 h 通过线性变换生成的对齐模式向量，其维度与特征向量相同。这里，激活函数 sdp 是一个指数线性单元（ELU）， W 和 b 分别是 s 函数的权重和偏置项， W 和 W 是权重参数， b 是偏置项。然后， $p = \frac{1}{4} p; p; \dots; p f g$ 表示所有样本的概率，第 i 个 EEG 样本的概率可以表示如下：

$$p = \frac{1}{4} P \frac{\exp z h}{i^{1/4} \exp z h}; \tag{7}$$

最后，通过扩展自注意力机制提取的特征，通过扩展自注意力机制提取的第 i 个注意力特征可以按如下方式计算：

TABLE 1
DEAP Database

Array name	Array shape	Array contents
Data	$40 \times 40 \times 8064$	video/trial $\times$ channel $\times$ data
Labels	40×4	video/trial $\times$ label

$$A_i = p_i \cdot h'_i. \quad (8)$$

In the last part of the proposed ACRNN, we employ the softmax layer as the classifier. The extracted spatiotemporal attentive features are $\mathbf{A} = \{A_1, A_2, \dots, A_n\}$, and the softmax classifier receives these extracted features as input to recognize emotion as follows:

$$P = \text{softmax}(W\mathbf{A} + b), \quad (9)$$

where $P = \{P_1, P_2, \dots, P_n\}$, P_i ($i = 1, 2, \dots, n$) represents the predicted probability of the i th EEG sample, and W and b are the weight and bias terms of the softmax function, respectively. Then, the cross-entropy error over all labeled samples is evaluated:

$$\mathcal{L} = - \sum_{i=1}^n \hat{Y}_i \log(P_i), \quad (10)$$

where $\hat{Y}_i$ is the label of the i th EEG sample, and the lower cross-entropy error $\mathcal{L}$ indicates higher emotion recognition accuracy.

In summary, we have designed a framework to extract features and classify emotion from raw EEG signals. We adopt a channel-wise attention mechanism to adaptively assign the weights of different channels, which can extract the intrinsic information among channels, then employ a CNN to extract the spatial information of encoded EEG signals. In addition, we adopt two-layer LSTM to explore the temporal information of different EEG samples, and we integrate the extended self-attention mechanism to assign weight to EEG samples based on the importance of each sample. Finally, spatiotemporal attentive features can be obtained for EEG emotion recognition.

4 EXPERIMENTS

Here, we introduce two widely used databases. Then, we introduce six deep learning methods and two traditional methods for comparison. We then demonstrate the model implementation in our experiments. Finally, we present and compare the experimental results obtained by the proposed method and compared methods.

4.1 Data Materials

To validate the performance of the proposed ACRNN, we conducted experiments on two widely used databases, i.e., the database for emotion analysis using physiological signals (DEAP) [2] and database for emotion recognition through EEG and ECG signals (DREAMER) [3]. The DEAP database includes the EEG and peripheral physiological signals of 32 participants recorded while they subjects watched 40 pieces of music videos. The database contains 32-channel

TABLE 2
DREAMER Database

Audio-visual stimuli	
Number of videos	18
Video content	Audio-Video
Video duration	65-393 s (M = 199 s)
Experiment information	
Number of participants	23
Number of males	14
Number of females	9
Age of participants	22-33
Rating scales	Arousal, Valence, Dominance
Rating values	1-5
Recorded signals	14-channel 128 Hz EEG

EEG signals and eight-channel peripheral physiological signals, where the EEG signals are used for emotional recognition, and the peripheral physiological signals are abnegated. In this experiment, the EEG signals were sampled at 512 Hz and then downsampled to 128 Hz. In addition, electrooculography (EOG) artifacts were removed using the blind source separation technique. The preprocessed EEG data of each trial contain 60-s trial data and 3-s baseline data. The emotional music videos include 40 one-minute clips, and the participants were asked to record their levels of arousal, valence, liking, and dominance for each video from 1 to 9. In our experiment, we selected valence and arousal as the emotional evaluation criteria, and the threshold to divide trials into two classes according to the rated levels of arousal and valence was set to five. Each subject file contained two arrays, and the data format of these files is detailed in Table 1.

The second database is DREAMER (Table 2), which is a multimodal database of EEG and ECG signals recorded during affect elicitation by means of audio visual stimuli. Signals from 23 participants (14 males and 9 females) were recorded, and participants were asked to record the levels of arousal, valence, and dominance after each stimuli. The EEG signals were recorded at a sampling rate of 128 Hz using an emotive EPOC system [49]. Each film clip is 65 to 393 s, which is sufficient to elicit single emotions. Moreover, the recorded EEG signals contain baseline signals and typically last 4 s before each film clip. In addition, most ocular artifacts (eye blinking, eye movement, cardiac interferences, etc.) have been removed with linear phase FIR filters. Furthermore, to avoid contaminating the data with multiple emotions, the recordings captured during the last 180 s of each clip were used for further analysis. The threshold of rating values is placed in the middle, where values less than or equal to 3 represent low valence, arousal, and dominance, and values greater than 3 represent high valence, arousal, and dominance.

4.2 Model Implementation

After the preprocessing stage, we obtained a total of EEG 800 samples for each subject in DEAP database, where each sample is S_i ($i = 1, 2, \dots, 800$) $\in \mathbb{R}^{32 \times 384}$. For DREAMER, we obtained a total of EEG 1250 samples for each subject, where each sample is S_i ($i = 1, 2, \dots, 1250$) $\in \mathbb{R}^{14 \times 384}$. We shuffled all samples from different trials for each subject.

TABLE 1
DEAP 数据库

数组名称	数组形状	数组内容	数据 40 40 8064 视频/试验/通道/数据	标签
40 4	视频/试验/标签			

$$A^{1/4} p h: \tag{8}$$

在所提出的 ACRNN 的最后一部分，我们使用 softmax 层作为分类器。提取的时空注意力特征是 $A = A_1, A_2, \dots, A_{\lfloor \frac{M}{4} \rfloor}$ ，softmax 分类器接收这些提取的特征作为输入来识别情绪，如下所示：

$$PP = \text{softmax}(W \cdot A + b); \tag{9}$$

其中 $PP = P_1, P_2, \dots, P_{\lfloor \frac{M}{4} \rfloor}$ ， $P(i)$ ($i = 1, 2, \dots, n$) 表示第 i 个脑电图样本的预测概率， W 和 b 分别是 softmax 函数的权重和偏置项。然后，评估所有标记样本的交叉熵误差：

$$L = \sum_{i=1}^N -Y_i \log P_i; \tag{10}$$

其中 Y_i 是第 i 个 EEG 样本的标签，下交叉熵误差 L 表示更高的情绪识别准确率。

总之，我们设计了一个从原始 EEG 信号中提取特征和分类情绪的框架。我们采用通道注意力机制来自适应分配不同通道的权重，这可以提取通道之间的内在信息，然后使用 CNN 提取编码 EEG 信号的空间信息。此外，我们采用双层 LSTM 来探索不同 EEG 样本的时间信息，并将扩展的自注意力机制集成到框架中，根据每个样本的重要性为 EEG 样本分配权重。最后，可以得到用于 EEG 情绪识别的时空注意力特征。

4 实验

这里，我们介绍了两个广泛使用的数据库。然后，我们介绍了六种深度学习方法以及两种传统方法用于比较。接着，我们展示了实验中的模型实现。最后，我们呈现并比较了所提出的方法和对比方法的实验结果。

4.1 数据材料

为了验证所提出的 ACRNN 的性能，我们在两个广泛使用的数据库上进行了实验，即用于生理信号情绪分析的数据库（DEAP）[2]和用于脑电图与心电图信号情绪识别的数据库（DREAMER）[3]。DEAP 数据库包括 32 名受试者在观看 40 段音乐视频时记录的脑电图和外围生理信号

表 2
DREAMER 数据库

视听刺激	视频数量
	18
视频内容	音频-视频
视频时长（M = 199 秒）	实验信息
	参与者数量
	23
男性数量	14
女性人数	9
参与者的年龄	22-33
评分量表	唤醒度、效价、主导性
评分值	1-5
记录信号	14 通道 128 Hz 脑电图

该数据库包含 32 通道脑电图（EEG）信号和 8 通道外周生理信号，其中 EEG 信号用于情绪识别，而外周生理信号被忽略。在本实验中，EEG 信号以 512 Hz 的采样率采集，然后降采样至 128 Hz。此外，使用盲源分离技术去除了眼电图（EOG）伪影。每个试验的预处理 EEG 数据包含 60 秒的试验数据和 3 秒的基线数据。情绪音乐视频包括 40 个一分钟的视频片段，要求参与者对每个视频的唤醒度、效价、喜好度和支配度进行 1 到 9 的评分。在我们的实验中，我们选择了效价和唤醒度作为情绪评估标准，并根据唤醒度和效价的评分水平将试验分为两类，设定的阈值为 5。每个受试者文件包含两个数组，这些文件的格式在表 1 中详细说明。

第二个数据库是 DREAMER（表 2），这是一个多模态数据库，记录了通过视听刺激引发情绪时采集的脑电图（EEG）和心电图（ECG）信号。记录了 23 名参与者（14 名男性，9 名女性）的信号，并要求参与者在每次刺激后记录唤醒度、效价和支配度的水平。脑电图信号使用情感化 EPOC 系统[49]以 128 Hz 的采样率进行记录。每个视频片段持续 65 到 393 秒，足以引发单一情绪。此外，记录的脑电图信号包含基线信号，并且在每个视频片段前通常持续 4 秒。此外，大多数眼动伪影（眨眼、眼球运动、心脏干扰等）已通过线性相位 FIR 滤波器去除。此外，为了避免数据被多种情绪污染，用于进一步分析的是每个片段最后 180 秒的记录。评分值的阈值设置在中间，其中值小于或等于 3 代表低效价、低唤醒度和低支配度，值大于 3 代表高效价、高唤醒度和高支配度。

4.2 模型实现

预处理阶段后，我们在 DEAP 数据库中为每个受试者获得了总共 800 个 EEG 样本，其中每个样本是 $S_{\delta i}^{1/4} 1; 2; \dots; 800 P^2 R$ 。对于 DREAMER，我们为每个受试者获得了总共 1250 个 EEG 样本，其中每个样本是 $S_{\delta i}^{1/4} 1; 2; \dots; 1250 P^2 R$ 。我们将每个受试者来自不同试验的所有样本进行了随机排序。

TABLE 3
Baseline Model and Attention-Based Models
for EEG Emotion Recognition

Model \ Component	channel-wise attention	CNN	LSTM network	self-attention
CNN-RNN	×	✓	✓	×
A-CNN-RNN	✓	✓	✓	×
CNN-RNN-A	×	✓	✓	✓
ACRNN	✓	✓	✓	✓

Then, we used 10-fold cross-validation to evaluate the performance of the proposed and baseline methods. The average performance of the 10-fold validation process was taken as the final experimental results. The model was implemented with the TensorFlow framework and trained on an NVIDIA TITAN Xp pascal GPU. In addition, the Adam optimizer was employed to minimize the cross-entropy loss function, and the network parameters were optimized with a learning rate of 10^{-4} , and the dropout regularization was set to 0.5. Batch normalization was adopted to achieve better performance during training. The size of the convolution kernel was $a \times b$, the height was set $a = 32$ for DEAP and $a = 14$ for DREAMER, and the width was $b = 40$. The number of kernels was $k = 40$ and the pooling size was 1×75 with a stride of 10. In addition, we set the dimension of the hidden state in LSTM to 64.

4.3 Results and Analysis

To validate the effectiveness of the proposed method, we conducted extensive experiments on two databases. To validate the performance of the attention mechanisms, we designed three models to demonstrate the influence of the channel-wise attention and extended self-attention mechanisms, including CNN + RNN (CNN-RNN), Channel-wise attention mechanism + CNN + RNN (A-CNN-RNN) and CNN + RNN + extended self-attention mechanism (CNN-RNN-A). Details for these models are shown in Table 3. The CNN-RNN model comprised a CNN and LSTM network, and was designed to validate the effectiveness of the baseline framework, which can extract emotional features from raw EEG signals using a cascade framework. The A-CNN-RNN model comprised the channel-wise attention mechanism, a CNN and LSTM network, and was designed to validate the effectiveness of the channel-wise attention mechanism for the baseline framework. The CNN-RNN-A model comprised a CNN, LSTM network and the extended self-attention, and was designed to validate the effectiveness of extended self-attention mechanisms for the baseline framework. In addition, we compared the proposed method with three recent deep learning methods: continuous convolutional neural network (Conti-CNN) [24], a graph convolutional neural network (GCNN) [25], and a convolutional recurrent attention model (CRAM) [48]. The Conti-CNN can combine the features of multiple bands to improve recognition accuracy [24], the GCNN can adopt different entropy (DE) feature as inputs, and use the spectral graph filtering to extract features and recognize emotion [25], and the CRAM can utilize a CNN to encode the high-level representation of EEG signals and a recurrent attention mechanism to explore the temporal dynamics [48]. In addition, we employed two traditional feature-based classifiers for

comparison, including support vector machine (SVM) and decision tree (DT) [24]. All methods were processed by the same preprocessing as ACRNN, i.e., baseline signal removal and sliding windows.

For the traditional classifiers, we used DE features as inputs [17], [18]. DE feature has the balance ability of discriminating EEG pattern between low and high frequency energy, which is typically used as frequency-domain features in EEG emotion recognition [14], [47], [50]. According to the literatures [18] and [51], the band-pass filter is applied to EEG signals to obtain the sub-band signals, which approximately follow a Gaussian distribution. Consequently, five sub-bands were defined: 1) delta (1-3 Hz); 2) theta (4-7 Hz); 3) alpha (8-13 Hz); 4) beta (14-30 Hz); and 5) gamma (31-50 Hz). Note that we extracted DE features from the laterer four sub-band signals because the higher-frequency band (approximately 30-100 Hz) is more suitable for EEG emotion recognition [13]. The final feature vector was a concatenation of features from all channels. For DEAP, the final feature vector was $4 \times 32 = 128$ dimensions, and each subject yielded 800 samples, where each sample $\mathbf{X}_i \in \mathbb{R}^{32 \times 128}$ ($i = 1, 2, \dots, 800$). For DREAMER, the final feature vector was $4 \times 14 = 56$ dimensions, and each subject yielded 1250 samples, where each sample $\mathbf{X}_i \in \mathbb{R}^{14 \times 56}$ ($i = 1, 2, \dots, 1250$).

In our work, we conducted the experiments on the same subject with the proposed ACRNN and compared methods for subject-dependent EEG emotion recognition. We divided the sample data into training sets and test sets, and then used 10-fold cross validation [52]. Typically, 10-fold cross validation divides data into 10 equal data subsets, and one subset is used as the test set, and the other nine subsets form the training set. This process was repeated 10 times. For the DEAP database, the number of training samples was 720, and the remaining 80 samples for each subject were used as test samples. For the DREAMER database, the numbers of training and test samples were 1125 and 125, respectively.

To further analyze the contribution of channel-wise attention, we performed experiments to compute the channel weights of EEG signals by channel-wise attention. Figs. 6 and 7 show the average channel weights in DEAP and DREAMER, respectively. There are 32 channels in DEAP dataset and 14 channels in DREAMER dataset. As shown, the channel weights of EEG signals in DEAP and DREAMER are both different in channel-wise attention mechanism. Fig. 6 shows the channel weights of FC5, P3, C4, P8 are obviously greater than other channels on two dimensions in DEAP database. Fig. 7 shows the channel weights of T8 and F8 are obviously greater than other channels on three dimensions in DREMAER database. The electrodes are placed according to the international 10-20 system in two databases, the relationship between greater weight channels and brain region is as shown in Table 4. The results are also consistent with some studies, which has demonstrated the EEG signals related to emotions are mostly distributed in the frontal lobe, the temporal lobe and parietal lobe [30], [31], [53]. The greater channel weight also indicates the given channels are more related to emotions and more important in the EEG signals.

Table 5 shows the average recognition accuracies of the proposed and compared methods on valence and arousal in the DEAP database. As can be seen, the average recognition

表 3 基准模型和基于注意力的模型
脑电图情绪识别

模型 \ 组件	通道注意力	CNN	LSTM 网络	自注意力
CNN-RNN		@	@	
A-CNN-RNN	@	@	@	
CNN-RNN-A		@	@	@
ACRNN	@	@	@	@

然后，我们使用 10 折交叉验证来评估所提出的方法和基线方法的性能。将 10 折验证过程的平均性能作为最终的实验结果。该模型使用 TensorFlow 框架实现，并在 NVIDIA TITAN Xp pascal GPU 上进行训练。此外，采用 Adam 优化器来最小化交叉熵损失函数，网络参数使用学习率为 10 进行优化，并设置 dropout 正则化率为 0.5。训练过程中采用批量归一化以获得更好的性能。卷积核的大小为 $a \times b$ ，高度设置为 DEAP 为 $a=32$ ，DREAMER 为 $a=14$ ，宽度为 $b=40$ 。卷积核的数量为 $k=40$ ，池化大小为 1×75 ，步长为 10。此外，我们将 LSTM 的隐藏状态维度设置为 64。

4.3 结果与分析

为验证所提方法的有效性，我们在两个数据库上进行了大量实验。为验证注意力机制的性能，我们设计了三个模型来展示通道注意力机制和扩展自注意力机制的影响，包括 CNN + RNN（CNN-RNN）、通道注意力机制 + CNN + RNN（A-CNN-RNN）和 CNN + RNN + 扩展自注意力机制（CNN-RNN-A）。这些模型的详细信息如表 3 所示。CNN-RNN 模型由一个 CNN 和 LSTM 网络组成，设计用于验证基线框架的有效性，该框架能够使用级联框架从原始脑电信号中提取情绪特征。A-CNN-RNN 模型由通道注意力机制、CNN 和 LSTM 网络组成，设计用于验证通道注意力机制对基线框架的有效性。CNN-RNN-A 模型由 CNN、LSTM 网络和扩展自注意力机制组成，设计用于验证扩展自注意力机制对基线框架的有效性。此外，我们将所提出的方法与三种最近提出的深度学习方法进行了比较：连续卷积神经网络（Conti-CNN）[24]、图卷积神经网络（GCNN）[25]和卷积循环注意力模型（CRAM）[48]。Conti-CNN 可以将多个频段的特征结合起来以提高识别准确率[24]，GCNN 可以采用不同的熵（DE）特征作为输入，并使用频谱图滤波来提取特征和识别情绪[25]，CRAM 可以利用卷积神经网络（CNN）对脑电图（EEG）信号的高级表示进行编码，并使用循环注意力机制来探索时间动态[48]。此外

我们采用了两种基于特征的传统分类器进行比较，包括支持向量机（SVM）和决策树（DT）[24]。所有方法都经过了与 ACRNN 相同的预处理，即基线信号去除和滑动窗口。

对于传统分类器，我们使用 DE 特征作为输入[17], [18]。DE 特征具有平衡区分低频和高频能量之间 EEG 模式的能力，通常在 EEG 情绪识别中作为频域特征使用[14], [47], [50]。根据文献[18]和[51]的描述，对 EEG 信号应用带通滤波器以获得子带信号，这些信号近似服从高斯分布。因此，定义了五个子带：1) delta (1-3 Hz); 2) theta (4-7 Hz); 3) alpha (8-13 Hz); 4) beta (14-30 Hz); 以及 5) gamma (31-50 Hz)。请注意，我们从后四个子带信号中提取 DE 特征，因为较高频带（约 30-100 Hz）更适合 EEG 情绪识别[13]。最终特征向量是所有通道特征的连接。对于 DEAP，最终特征向量为 $4 \times 32 = 128$ 维，每个受试者产生 800 个样本，其中每个样本 $X \in R(i = 1, 2, \dots, 800)$ 。对于 DREAMER，最终特征向量为 $4 \times 14 = 56$ 维，每个受试者产生 1250 个样本，其中每个样本 $X \in R(i = 1, 2, \dots, 1250)$ 。

在我们的工作中，我们使用所提出的 ACRNN 对同一受试者进行了实验，并与适用于受试者依赖型脑电图情绪识别的方法进行了比较。我们将样本数据分为训练集和测试集，然后使用了 10 折交叉验证[52]。通常，10 折交叉验证将数据分为 10 个相等的数据子集，其中一个子集用作测试集，其余 9 个子集构成训练集。这个过程重复了 10 次。对于 DEAP 数据库，训练样本数量为 720，每个受试者剩余的 80 个样本用作测试样本。对于 DREAMER 数据库，训练样本和测试样本的数量分别为 1125 和 125。

为了进一步分析通道注意力机制的贡献，我们进行了实验来计算 EEG 信号的通道权重。图 6 和图 7 分别展示了 DEAP 和 DREAMER 数据集中平均的通道权重。DEAP 数据集有 32 个通道，DREAMER 数据集有 14 个通道。如图所示，DEAP 和 DREAMER 中的 EEG 信号在通道注意力机制下具有不同的通道权重。图 6 显示，在 DEAP 数据库中，FC5、P3、C4 和 P8 的通道权重在两个维度上明显大于其他通道。图 7 显示，在 DREMAER 数据库中，T8 和 F8 的通道权重在三个维度上明显大于其他通道。两个数据库中的电极按照国际 10-20 系统放置，权重较大的通道与脑区的对应关系如表 4 所示。这些结果也与一些研究一致，这些研究表明与情绪相关的 EEG 信号主要分布在额叶、颞叶和顶叶[30]、[31]、[53]。更大的通道权重也表明给定通道与情绪更相关，在脑电图信号中更重要。

表 5 显示了在 DEAP 数据库中，所提出的方法与对比方法在效价和唤醒度上的平均识别准确率。可以看出，平均识别

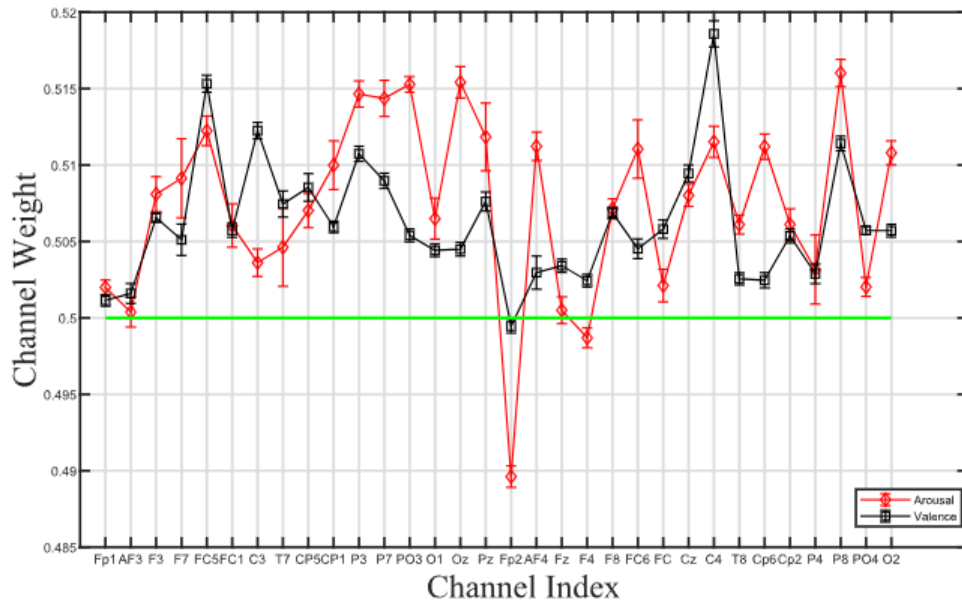


Fig. 6. The average channel weight on DEAP database.

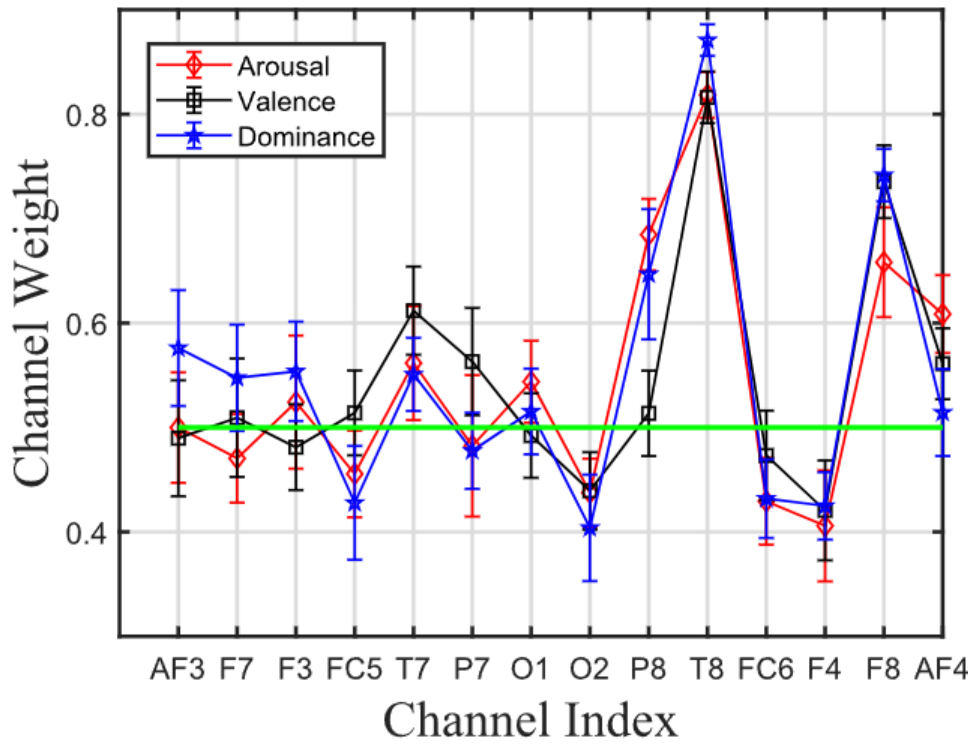


Fig. 7. The average channel weight on DREAMER database.

accuracy of A-CNN-RNN improved by approximately 30 and 25 percent compared to the baseline framework CNN-RNN on two dimensions because channel-wise attention focuses on the spatial features among different channels. In addition, the proposed ACRNN improved the average recognition accuracy by 0.6 and 0.5 percent compared to A-CNN-RNN because the ACRNN combines the channel-wise and extended self-attention mechanisms to extract the spatiotemporal attentive information of the EEG signals. In addition, we found that CNN-RNN-A improved recognition accuracy by approximately 27 percent on two dimensions compared to CNN-RNN because the extended self-attention mechanism of CNN-RNN-A extracts attentive information according to the importance of each sample. In addition, the proposed ACRNN improved recognition accuracy by approximately 3 percent on two dimensions compared to CNN-RNN-A because the proposed model exploits both attention mechanisms simultaneously. Compared to the three deep learning methods (Conti-CNN, CRAM, and GCNN), the experimental results indicate that the proposed ACRNN improved the average recognition accuracy by 10, 8 and 5 percent, respectively. Compared to traditional methods, the ACRNN achieved better recognition performance than the traditional methods, e.g., DT and the SVM.

Table 6 shows the average recognition accuracies of the compared methods obtained on the DREAMER database. As shown, emotion recognition accuracy improved significantly

TABLE 4
The Relationship Between Greater Weight Channels and Brain Region in DEAP and DREAMER

Database	Channel	Brain region
DEAP	F3	Frontal
	FC5	Frontal
	P3	Parietal
	P7	Parietal
	F8	Frontal
	Cz	Central
	C4	Central
	P8	Parietal
DREAMER	T8	Temporal
	F8	Frontal

TABLE 5
Average Accuracies (%) of Different Methods on the Valence and Arousal Classification Tasks of DEAP Database (MEAN $\pm$ STD. DEV.)

	Valence	Arousal
DT	75.95 $\pm$ 4.76	78.18 $\pm$ 5.45
SVM	89.33 $\pm$ 7.41	89.99 $\pm$ 6.74
Conti-CNN	82.77 $\pm$ 4.47	81.55 $\pm$ 6.55
CRAM	87.09 $\pm$ 7.49	84.46 $\pm$ 9.27
GCNN	88.24 $\pm$ 3.18	87.72 $\pm$ 3.32
CNN-RNN	62.75 $\pm$ 7.53	67.12 $\pm$ 9.13
A-CNN-RNN	91.48 $\pm$ 5.02	91.59 $\pm$ 5.42
CNN-RNN-A	89.15 $\pm$ 6.66	89.96 $\pm$ 5.93
ACRNN	93.72 $\pm$ 3.21	93.38 $\pm$ 3.73

TABLE 6
Average Accuracies (%) of Different Methods on the Valence, Arousal and Dominance Classification Tasks of DREAMER Database (MEAN $\pm$ STD. DEVC.)

	Valence	Arousal	dominance
DT	68.81 $\pm$ 6.87	67.50 $\pm$ 7.28	67.43 $\pm$ 6.73
SVM	76.71 $\pm$ 5.89	77.54 $\pm$ 5.62	75.76 $\pm$ 5.63
Conti-CNN	81.72 $\pm$ 5.24	82.48 $\pm$ 5.11	82.58 $\pm$ 5.28
CRAM	92.27 $\pm$ 2.95	93.03 $\pm$ 1.87	93.34 $\pm$ 1.78
GCNN	88.87 $\pm$ 3.58	88.79 $\pm$ 3.86	88.54 $\pm$ 3.89
CNN-RNN	78.59 $\pm$ 13.87	77.66 $\pm$ 13.34	77.75 $\pm$ 14.22
A-CNN-RNN	97.47 $\pm$ 2.32	97.92 $\pm$ 1.60	98.15 $\pm$ 1.76
CNN-RNN-A	96.61 $\pm$ 3.42	97.36 $\pm$ 2.63	97.54 $\pm$ 2.16
ACRNN	97.93 $\pm$ 1.73	97.98 $\pm$ 1.92	98.23 $\pm$ 1.42

on the DREAMER database. For example, the proposed ACRNN improved the average recognition accuracy by 15, 9 and 5 percent compared to Conti-CNN, GCNN and CRAM, respectively. Thus, the proposed ACRNN can achieve the best recognition accuracy among all compared methods, and the experimental results demonstrate the effectiveness of integrating the two attention mechanisms into the CNN-RNN.

To demonstrate the performance of the proposed method and compared methods for each subject, we conducted experiments on each subject. Figs. 8, 9, 10, 11, and 12 show the average accuracy and standard deviation of each subject on each dimension. As can be seen, the traditional SVM and DT methods achieved good average recognition accuracy

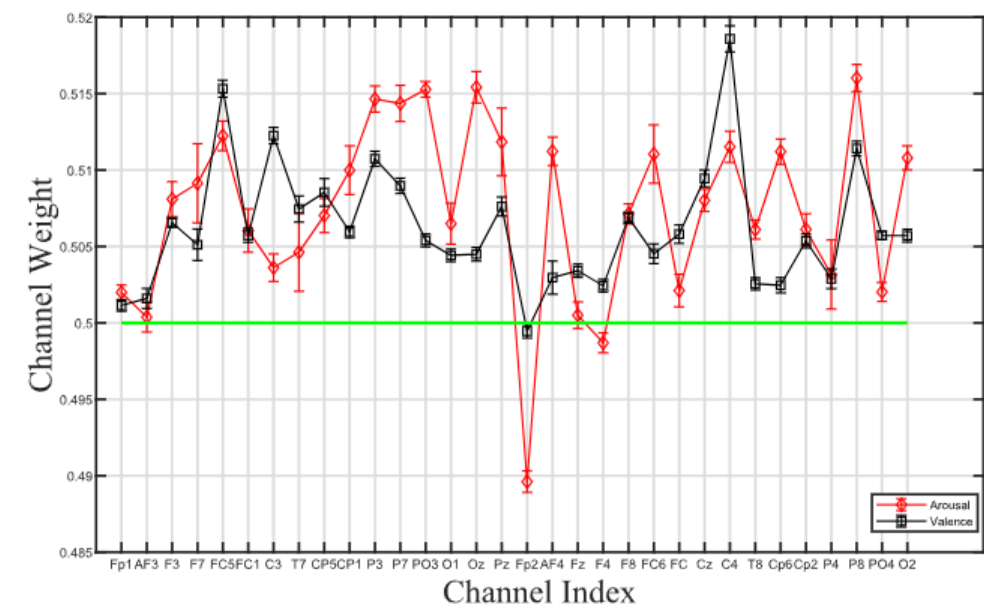


图 6. DEAP 数据库上的平均通道权重。

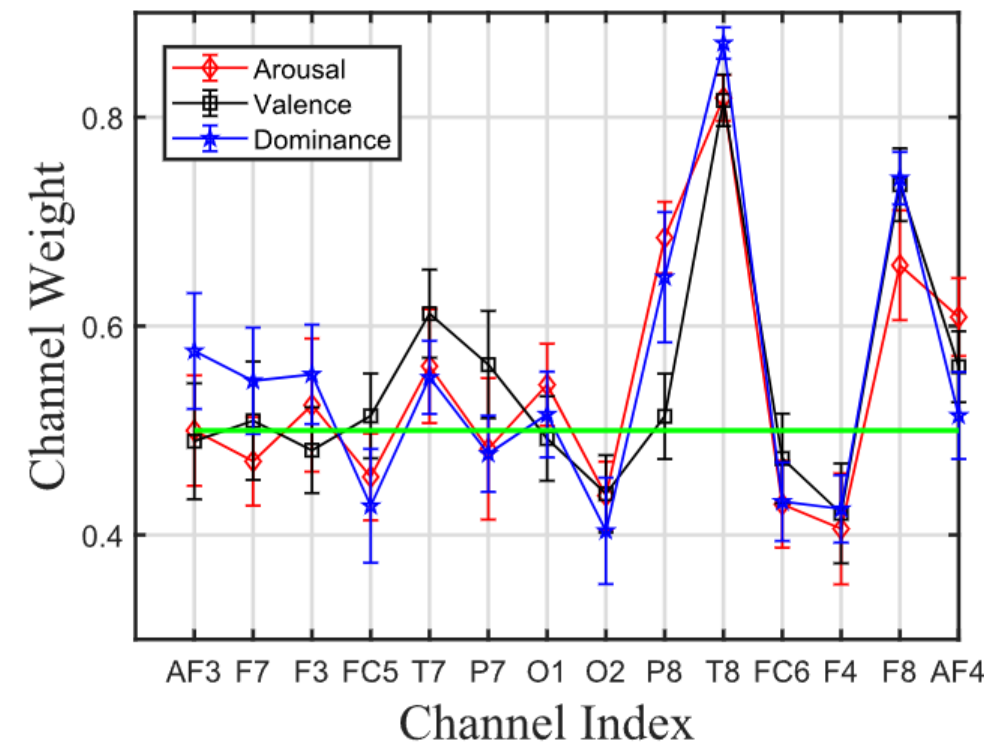


图 7. DREAMER 数据库上的平均通道权重。

与基线框架 CNNRNN 相比，A-CNN-RNN 在两个维度上的准确率提高了约 30%和 25%，因为通道注意力机制专注于不同通道之间的空间特征。此外，所提出的 ACRNN 与 ACNN-RNN 相比，平均识别准确率提高了 0.6%和 0.5%，因为 ACRNN 结合了通道注意力和扩展自注意力机制来提取脑电信号的时空注意力信息。此外，我们发现 CNN-RNN-A 在两个维度上比 CNN-RNN 的识别准确率提高了约 27%，因为 CNN-RNN-A 的扩展自注意力机制根据每个样本的重要性提取注意力信息。此外，所提出的 ACRNN 在两个维度上比 CNN-RNN-A 的识别准确率提高了约 3%，因为该模型同时利用了两种注意力机制。与三种深度学习方法（Conti-CNN、CRAM 和 GCNN）相比，实验结果表明，所提出的 ACRNN 分别提高了平均识别准确率 10%、8%和 5%。与传统方法相比，ACRNN 在识别性能上优于传统方法，例如 DT 和 SVM。

表 6 显示了在 DREAMER 数据库上获得的比较方法的平均识别准确率。如图所示

表 4 更大权重通道之间的关系

以及脑区在 DEAP 和 DREAMER

数据库	通道	脑区
DEAP	F3	额叶
	FC5	额叶
	P3	顶叶
	P7	顶叶
	F8	额叶
	Cz	中央
	C4	中央
	P8	顶叶
DREAMER	T8	颞叶
	F8	额叶

表 5 DEAP 数据库在效价和唤醒分类任务上不同方法的平均准确率 (%)

(均值 标准差)			
效价		唤醒	
DT	75.95 4.76	78.18 5.45	
SVM	89.33 7.41	89.99 6.74	
Conti-CNN	82.77 4.47	81.55 6.55	
CRAM	87.09 7.49	84.46 9.27	
GCNN	88.24 3.18	87.72 3.32	
CNN-RNN	62.75 7.53	67.12 9.13	
A-CNN-RNN	91.48 5.02	91.59 5.42	CNN-RNN-A 89.15 6.66 89.96 5.93
ACRNN	93.72 3.21	93.38 3.73	

表 6 DREAMER 情感效价、唤醒度和支配度分类任务中不同方法的平均准确率 (%)

数据库（均值 ± 标准差）											
效价				唤醒度				主导性			
DT	68.81	6.87		67.50	7.28		67.43	6.73			
SVM	76.71	5.89	77.54	5.62	75.76	5.63	Conti-CNN	81.72	5.24	82.48	
	5.11	82.58	5.28	CRAM	92.27	2.95	93.03	1.87	93.34	1.78	GCNN
	88.87	3.58	88.79	3.86	88.54	3.89	CNN-RNN	78.59	13.87	77.66	13.34
	77.75	14.22	A-CNN-RNN	97.47	2.32	97.92	1.60	98.15	1.76	CNN-	
	RNN-A	96.61	3.42	97.36	2.63	97.54	2.16	ACRNN	97.93	1.73	97.98
	1.92	98.23	1.42								

在 DREAMER 数据库上，情绪识别准确率显著提高。例如，与 Conti-CNN、GCNN 和 CRAM 相比，所提出的 ACRNN 分别将平均识别准确率提高了 15%、9%和 5%。因此，所提出的 ACRNN 在所有比较方法中实现了最佳识别准确率，实验结果证明了将两种注意力机制集成到 CNN-RNN 中的有效性。

为了展示所提出的方法和比较方法在每个受试者上的性能，我们对每个受试者进行了实验。图 8、9、10、11 和 12 显示了每个受试者在每个维度上的平均准确率和标准差。如图所示，传统的 SVM 和 DT 方法实现了良好的平均识别准确率

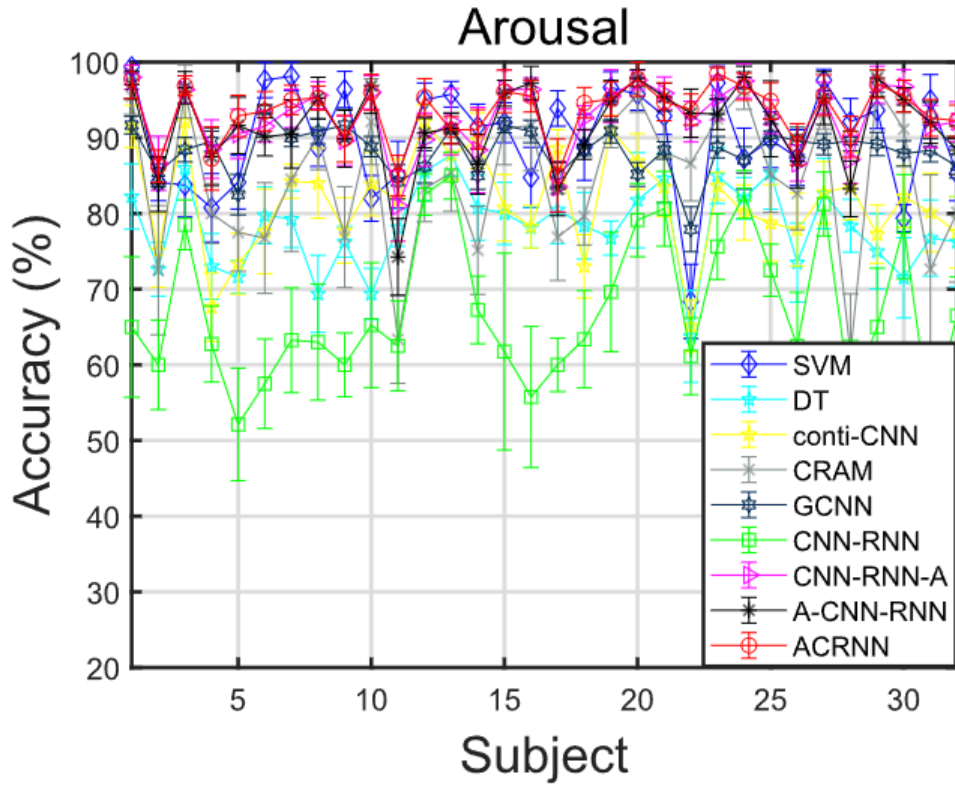


Fig. 8. Average accuracies (%) on each subject of different methods on arousal classification tasks on DEAP database.

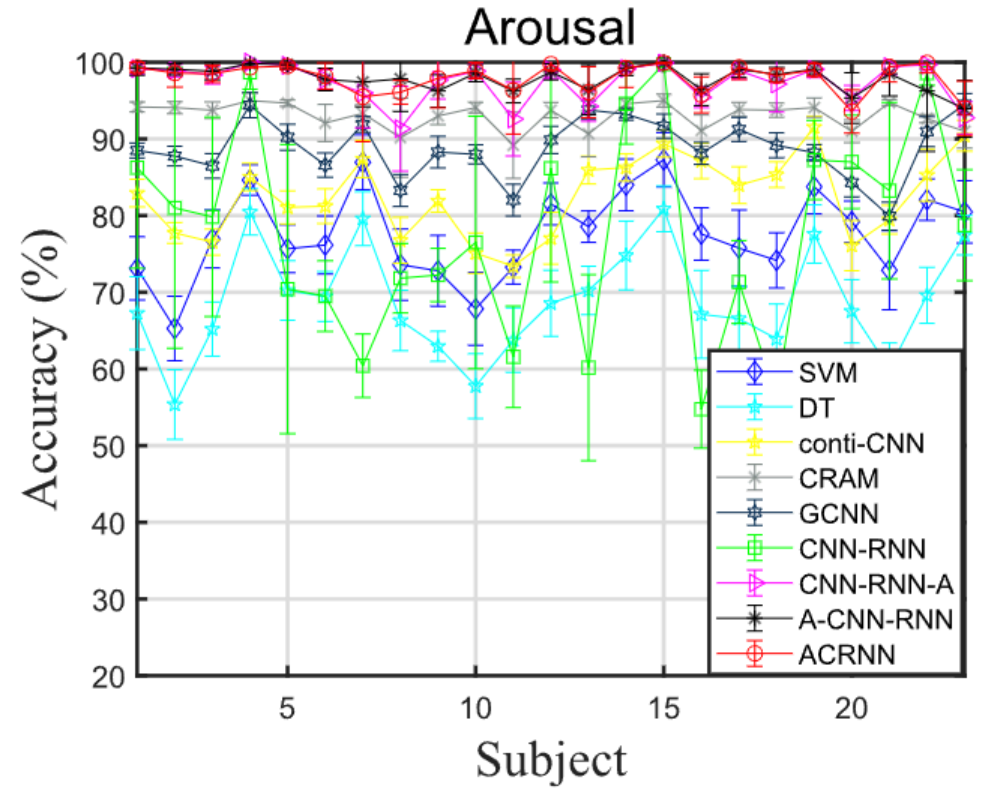


Fig. 10. Average accuracies (%) on each subject of different methods on arousal classification tasks on DREAMER database.

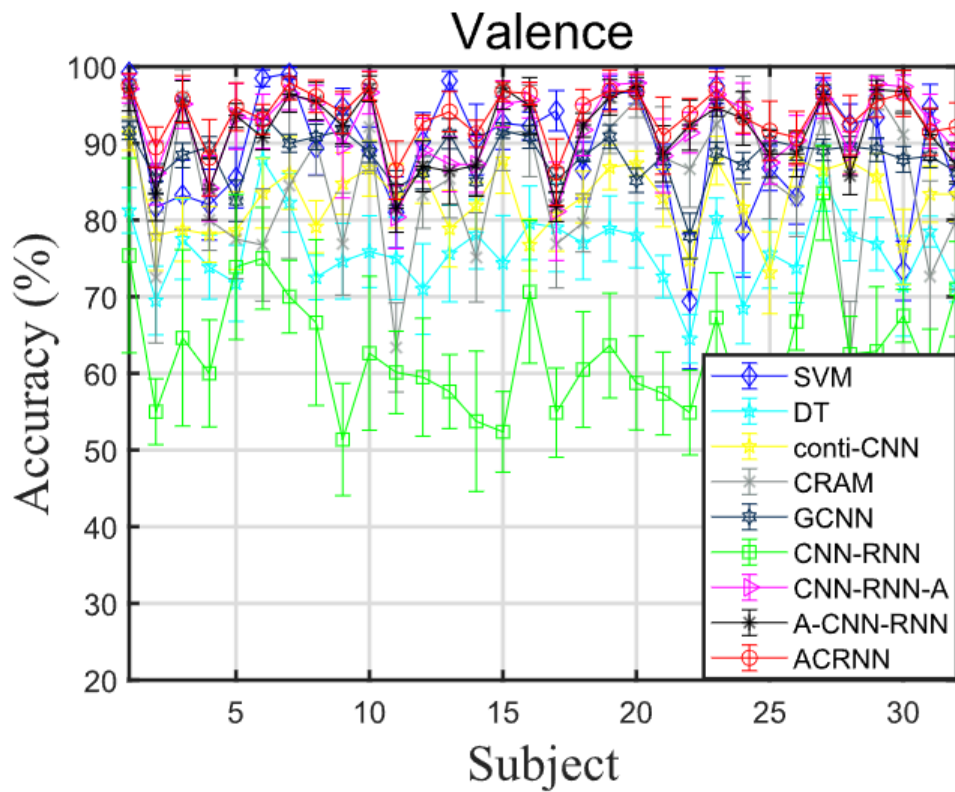


Fig. 9. Average accuracies (%) on each subject of different methods on valence classification tasks on DEAP database.

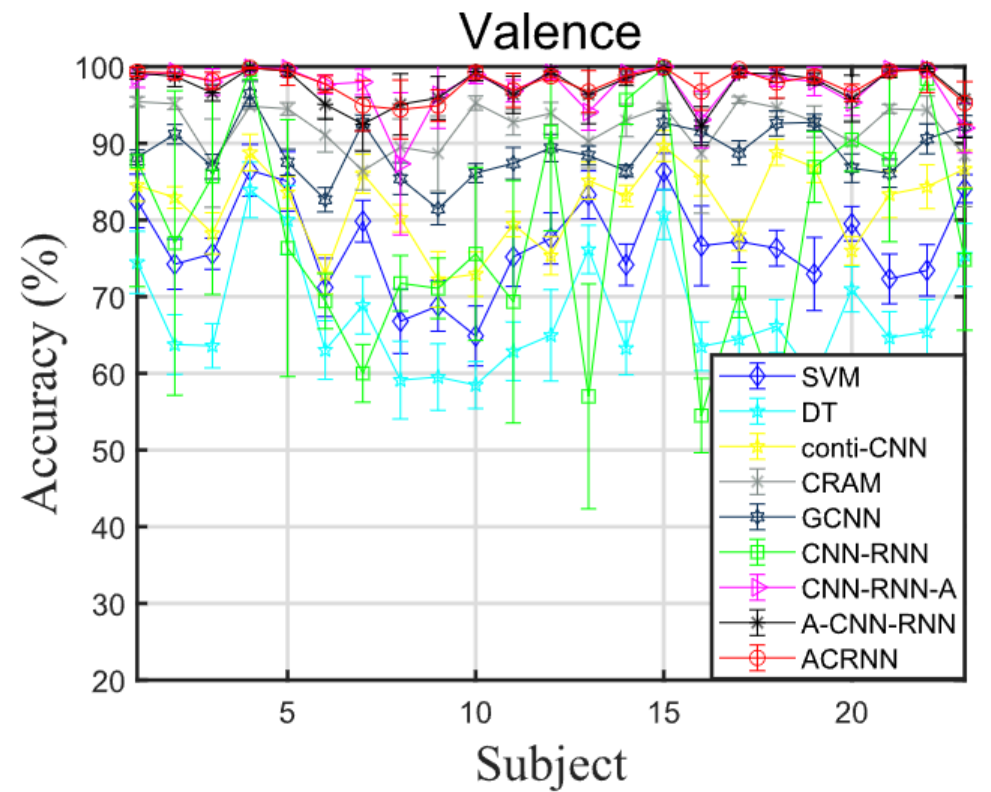


Fig. 11. Average accuracies (%) on each subject of different methods on valence classification tasks on DREAMER database.

on some subjects, however, the standard deviations were very large. In addition, the compared methods performed worse on some subjects. However, we found that the three attention-based methods achieved better average recognition accuracy on each subject, and the standard deviations were less than those of the compared methods. Thus, the experimental results demonstrate that attention-based methods can work better than the compared methods for each subject. Furthermore, the results indicate that the proposed ACRNN combines channel-wise attention module and extended self-attention to exploit more discriminative information for EEG emotion recognition and can achieve superior recognition accuracy on two public databases.

5 DISCUSSIONS

EEG-based emotion recognition is widely used to help computers better understand the current emotional state of the user [21]. Traditional EEG recognition methods first design hand-

crafted features from raw EEG signals, and then employ classifier to classify these features. Recently, deep learning methods have an employed end-to-end technique to recognize emotion from raw EEG signals. However, it still remains challenging to extract more discriminative features from raw signals for EEG emotion recognition. Thus, a discriminative framework to recognize the emotional state from the raw EEG signals is required. Considering that raw EEG signals contain spatial information by the intrinsic relationship among different channels and time dependencies among temporal slices, we have proposed ACRNN to extract spatial and temporal attentive information and classify the emotional state of a subject.

In this framework, a channel-wise attention mechanism extracts the difference among channels from the EEG signals by assigning the weights to different channels, and a CNN is designed to extract the feature map as spatial information by the convolution operation on all channels of the EEG signals. Different from some methods based on brain cognitive function, they need to focus on EEG channels and

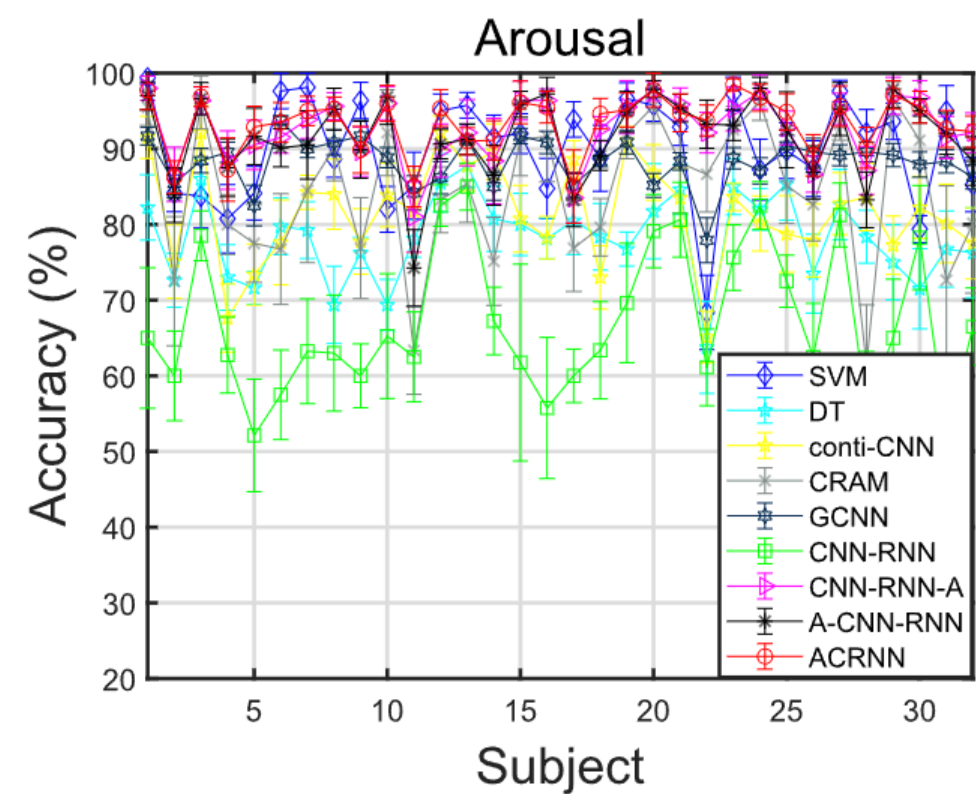


图 8. DEAP 数据库上不同方法在唤醒分类任务中每个受试者的平均准确率(%)

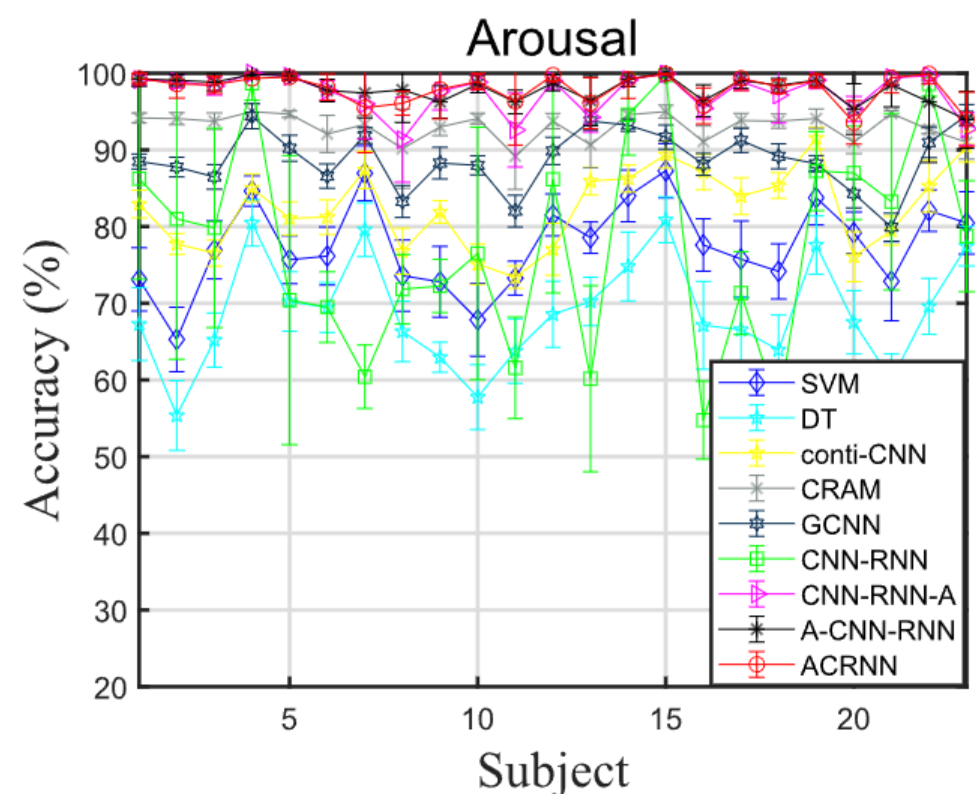


图 10. DREAMER 数据库上不同方法在唤醒分类任务中每个受试者的平均准确率(%)

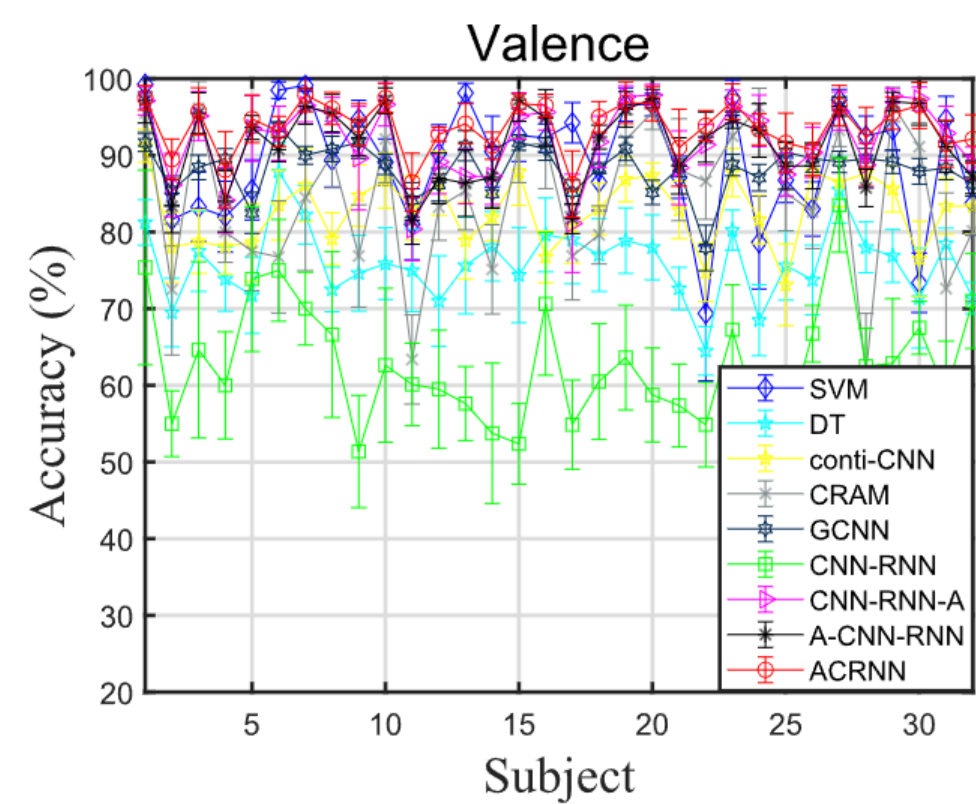


图 9. DEAP 数据库上不同方法在效价分类任务中每个受试者的平均准确率(%)

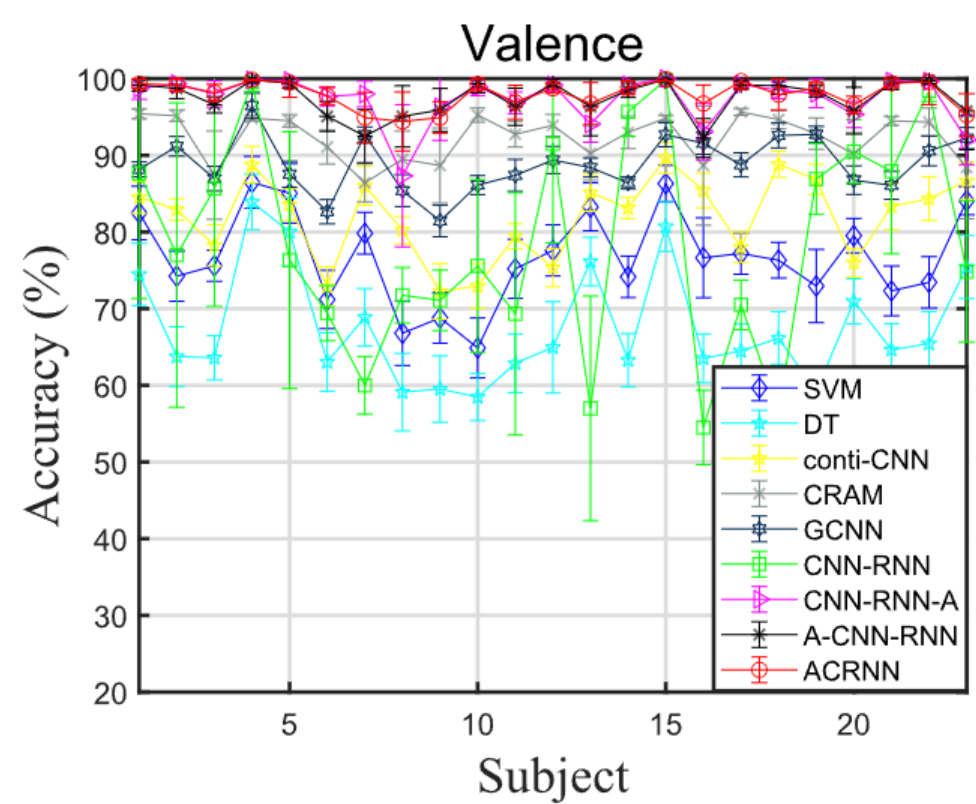


图 11. DREAMER 数据库上不同方法在效价分类任务中每个受试者的平均准确率(%)

然而，在一些受试者上，标准差非常大。此外，与其他方法相比，一些受试者的表现更差。但是，我们发现基于注意力的三种方法在每个受试者上实现了更高的平均识别准确率，且标准差小于其他方法。因此，实验结果表明，基于注意力的方法在每个受试者上表现优于其他方法。此外，结果还表明，所提出的 ACRNN 结合了通道注意力模块和扩展自注意力机制，能够为脑电情感识别提取更多判别性信息，并在两个公共数据集上实现了更高的识别准确率。

5 讨论

基于脑电的情绪识别被广泛应用于帮助计算机更好地理解用户的当前情绪状态 [21]

传统的脑电识别方法首先从原始脑电信号中设计手工特征，然后使用分类器对这些特征进行分类。近年来，深度学习方法采用了端到端技术从原始脑电信号中识别情绪。然而，从原始信号中提取更具判别性的特征用于脑电情绪识别仍然具有挑战性。因此，需要一个判别性框架来从原始脑电信号中识别情绪状态。考虑到原始脑电信号通过不同通道之间的内在关系和时间切片之间的时间依赖性包含空间信息，我们提出了 ACRNN 来提取空间和时间注意力信息，并对受试者的情绪状态进行分类。

在这个框架中，通道注意力机制通过为不同通道分配权重来从脑电图信号中提取通道之间的差异，并设计了一个卷积神经网络，通过在脑电图信号的所有通道上执行卷积操作来提取特征图作为空间信息。与一些基于脑认知功能的方法不同，它们需要关注脑电图通道。

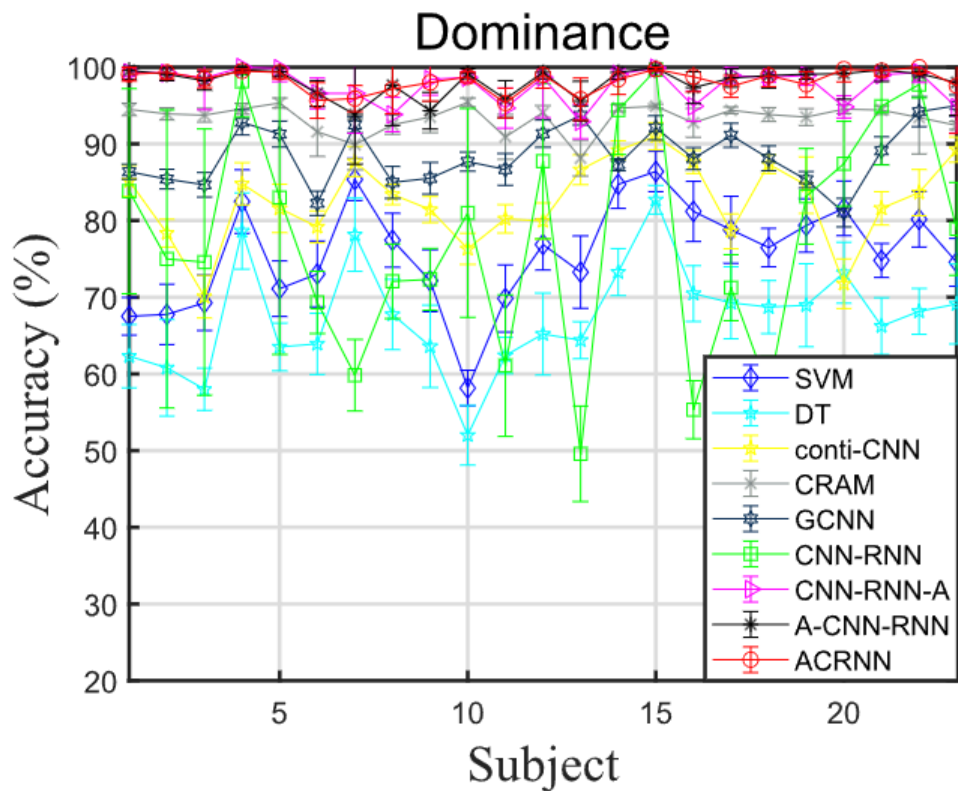


Fig. 12. Average accuracies (%) on each subject of different methods on dominance classification tasks on DREAMER database.

to design relevant features [54], [55], [56]. For example, Li *et al.* constructed emotion-related brain networks with phase locking value (PLV) and adopted a multiple feature fusion approach to combine the compensative activation and connection information for emotion recognition [55], Wang *et al.* combined brain directed connectivity (BDC) and DE features which are in different frequency bands among brain areas to extract discriminative information to improve recognition accuracy [56]. However, our proposed method adopt channel-wise attention to allocate weights in different channels. To further analyze the importance of different channels, we compute the average channel weights. From Figs. 6 and 7, we can find the channel weights of EEG signals in DEAP and DREAMER appear different according to the channel-wise attention mechanism. The channel weights of FC5, P3, C4, P8 are clearly greater than the other channels on two dimensions in DEAP database, and the channel weights of T8 and F8 are obviously greater than the other channels on three dimensions in DREMAER database. The result demonstrates that the EEG signals relevant to emotions are mostly distributed in the frontal lobe, the temporal lobe and the parietal lobe, which is consistent with the existing studies [30], [31], [53]. It can also be seen that the channels with computed greater weights are more related to emotions and thus more important in EEG-based emotion recognition. To demonstrate the effectiveness of the channel-wise attention mechanism, we integrate channel-wise attention into the baseline CNN-RNN framework, and the experimental results demonstrate that the channel-wise attention of the A-CNN-RNN can improve the average accuracy by approximately 30 percent compared to the CNN-RNN model on the DEAP and DREAMER databases because the channel-wise attention mechanism can transform channels to a probability distribution as weights and recode the EEG signals based on the transformed weights. In addition, the extended self-attention mechanism is designed to explore the importance of different EEG samples. To demonstrate the effectiveness of the extended self-attention mechanism, we integrated it into the baseline CNN-RNN framework, and the experimental results show that extended self-attention can improve average accuracy by 27

and 29 percent compared to the CNN-RNN model on these databases. The experimental results also demonstrate that the extended self-attention mechanism focuses on more important EEG samples by scoring the probability based on the similarities among samples.

In summary, the channel-wise attention and extended self-attention mechanisms improve the average accuracy greater than 25 percent on two both databases. This indicates that these attention mechanisms can improve EEG emotion recognition and achieve comparable recognition results. However, compared to extended self-attention, channel-wise attention improved the average recognition accuracy by approximately 2 and 1 percent on the DEAP and DREAMER databases, respectively. This indicates that channel-wise attention performs slightly better than extended self-attention. In summary, the proposed ACRNN is a cascade framework that integrates channel-wise attention and extended self-attention mechanisms. It can effectively extract spatiotemporal attentive features simultaneously. In addition, our all experimental results are obtained by 10-fold cross-validation, the high accuracy and low standard deviation have also demonstrated that the proposed ACRNN can achieve superior recognition accuracy.

6 CONCLUSION

In this paper, we have proposed an end-to-end deep learning method for EEG emotion recognition. The proposed ACRNN takes the spatial information, temporal information and attentive information of EEG signals into consideration. In addition, we integrate channel-wise attention into a CNN, which can extract the attentive information among the channels, and further extract spatial attentive features. We also integrated extended self-attention into RNN, which can extract attentive information based on the importance of each sample. Finally, extensive experimental results have demonstrated that the proposed ACRNN achieved average accuracies of 93.72 and 93.38 percent on the valence and arousal classification tasks in the DEAP database, respectively. In addition, the proposed ACRNN achieved average accuracies of 97.93, 97.78 and 98.23 percent on valence, arousal, and dominance classification tasks in the DREAMER database, respectively. Compared to existing methods, it is clear that the proposed ACRNN improved EEG emotion recognition accuracy in the DEAP and DREAMER databases. In future work, we will study the trial-based and inter-subjects EEG emotion recognition based on the attention mechanism.

ACKNOWLEDGMENTS

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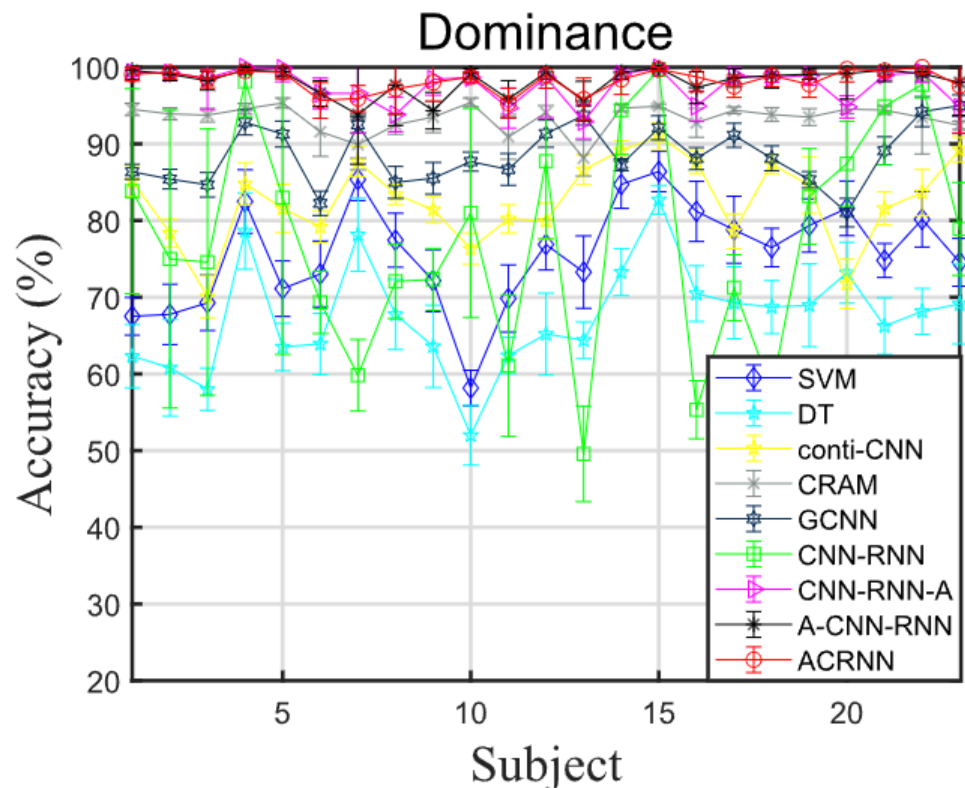


图 12. 在 DREAMER 数据库上，不同方法在主导分类任务中对每个被试的平均准确率 (%)

为了设计相关特征[54], [55], [56], 例如, Li 等人利用相位锁定值 (PLV) 构建了与情绪相关的脑网络, 并采用多特征融合方法来结合补偿激活和连接信息以进行情绪识别[55], Wang 等人结合了不同脑区频率带中的脑定向连接 (BDC) 和 DE 特征来提取判别信息以提高识别准确率[56]。然而, 我们提出的方法采用通道注意力机制来分配不同通道的权重。为了进一步分析不同通道的重要性, 我们计算了平均通道权重。从图 6 和图 7 中, 我们可以发现根据通道注意力机制, DEAP 和 DREAMER 数据库中 EEG 信号的通道权重有所不同。在 DEAP 数据库中, FC5、P3、C4、P8 的通道权重在两个维度上明显大于其他通道, 而在 DREAMER 数据库中, T8 和 F8 的通道权重在三个维度上明显大于其他通道。结果表明, 与情绪相关的脑电信号主要分布在额叶、颞叶和顶叶, 这与现有研究[30]、[31]、[53]的结果一致。同时可以看出, 计算得到的权重较大的通道与情绪的相关性更高, 因此在脑电信号情绪识别中更为重要。为了展示通道注意力机制的有效性, 我们将通道注意力机制整合到基准 CNN-RNN 框架中, 实验结果表明, 与 CNN-RNN 模型相比, A-CNN-RNN 的通道注意力机制在 DEAP 和 DREAMER 数据库上平均准确率提高了约 30%, 因为通道注意力机制可以将通道转换为权重概率分布, 并根据转换后的权重重新编码脑电信号。此外, 我们还设计了扩展的自注意力机制来探索不同脑电样本的重要性。为了展示扩展自注意力机制的有效性, 我们将其整合到基准 CNN-RNN 框架中。

实验结果表明, 与 CNN-RNN 模型相比, 扩展自注意力机制在这些数据库上可以提高平均准确率 27%和 29%。实验结果还表明, 扩展自注意力机制通过基于样本间相似性打分, 专注于更重要的脑电图样本。

总之, 通道注意力机制和扩展自注意力机制在两个数据库上均将平均准确率提高了 25%以上。这表明这些注意力机制能够提升脑电图情绪识别能力, 并实现相当不错的识别结果。然而, 与扩展自注意力机制相比, 通道注意力机制在 DEAP 和 DREAMER 数据库上分别将平均识别准确率提高了约 2%和 1%。这表明通道注意力机制的表现略优于扩展自注意力机制。总之, 所提出的 ACRNN 是一种级联框架, 集成了通道注意力机制和扩展自注意力机制。它能够有效同时提取时空注意力特征。此外, 我们所有实验结果均通过 10 折交叉验证获得, 高准确率和低标准差也证明所提出的 ACRNN 能够实现更优的识别准确率。

6 结论

在这篇论文中, 我们提出了一种用于脑电图情绪识别的端到端深度学习方法。我们提出的 ACRNN 考虑了脑电图信号的空间信息、时间信息和注意力信息。此外, 我们将通道注意力集成到 CNN 中, 这可以提取通道之间的注意力信息, 并进一步提取空间注意力特征。我们还将扩展自注意力集成到 RNN 中, 这可以根据每个样本的重要性提取注意力信息。最后, 大量的实验结果表明, 我们提出的 ACRNN 在 DEAP 数据库的效价和唤醒度分类任务上分别达到了 93.72%和 93.38%的平均准确率。此外, 在 DREAMER 数据库的效价、唤醒度和支配度分类任务上, ACRNN 分别达到了 97.93%、97.78%和 98.23%的平均准确率。与现有方法相比, 很明显我们提出的 ACRNN 提高了 DEAP 和 DREAMER 数据库中的脑电图情绪识别准确率。在未来的工作中, 我们将基于注意力机制研究基于试验和跨被试的脑电图情绪识别。

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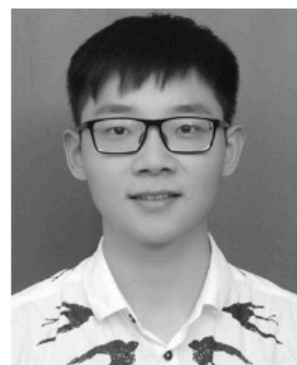
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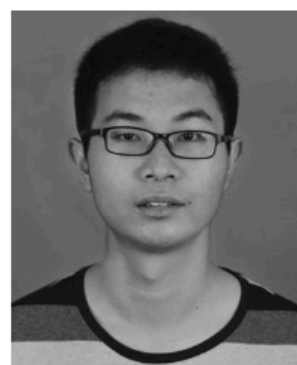
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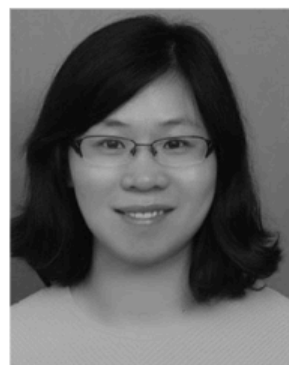
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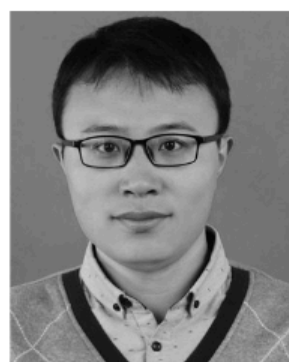
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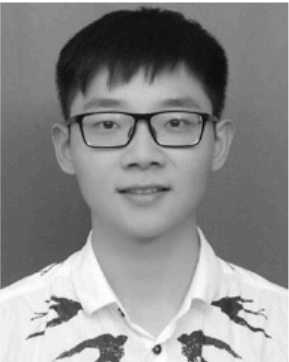
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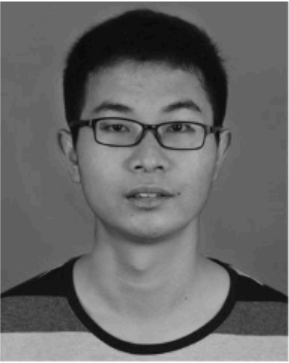
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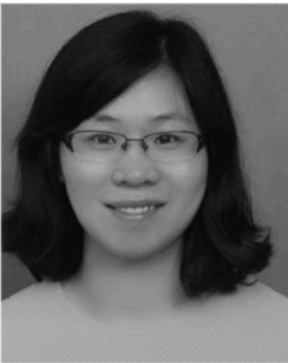
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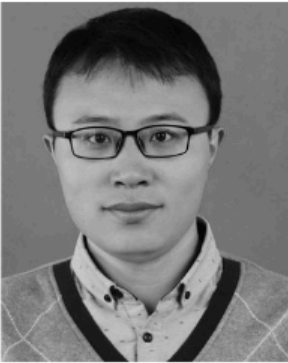
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